Beating Monte Carlo on rare circuit outputs: envelopes, proof-carrying sampling, and their limits

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Abstract

Given a Boolean circuit C and a product distribution D, how can one estimate the tiny probability $p = \Pr_D[C(x) = 1]$ faster than Monte Carlo, whose cost grows like $1/p$, without ever being wrong? We study procedures that are sound on *every* circuit (their confidence intervals are valid with probability $1 - \delta$) and efficient on as wide a class as possible, first without help and then with an unbounded prover who sends a single argument to a randomized polynomial-time verifier. The organising principle is that *upper bounds need deduction, lower bounds need proposals, and sampling pays for the surprise that is left.* An *envelope*, a tractable $\varphi \geq \mathbf{1}_C$ with computable mass, turns a deduction into an importance sampler with bounded weights whose cost is the looseness $K = \mathbb{E}_D\varphi/p$. Without a prover, a Decompose–Force–Sample–Branch procedure is sound everywhere and costs $\mathrm{poly} \cdot K$ on every circuit whose ideal K-explanation tree is polynomial: products and read-once compositions of dense sub-circuits, backdoor triggers, small cutsets, DNFs and DNNFs, and their locally disguised versions. Every circuit of logarithmic *context width* compiles to a polynomial-size DNNF and so has an FPRAS (covering, e.g., ORs of bounded-pathwidth CNFs, whose exact counting is #P-hard). For unate circuits the presumption of independence is certified by Harris–Janson sandwiches (to second order, with a matching barrier for pairwise data), and annealing becomes sound with exact Las Vegas samplers whose truncation is certified empirically. Further modules cover bounded treewidth and thresholds of independent blocks (exponential tilting, $K = O(\sqrt{k})$). With a prover, proof-carrying importance sampling, checking implications in any Cook–Reckhow proof system, is sound against every argument and complete for circuits whose rarity has a short proof of confinement to a tractable region; its cost is $\mathrm{poly}(|\pi|) + 2^{\sigma_u}$ with σ_u the unexplained surprise, and each bit of argument buys at most one bit of surprise. We prove the limits: no procedure beats sampling on all circuits (unless $\mathsf{NP} \subseteq \mathsf{BPP}$, and by more than $p^{-o(1)}$ under exponentially hard PRGs); verifiers that only evaluate C cannot certify upper bounds below $1/T$; no argument system is complete for upper bounds unless $\mathsf{coNP} \subseteq \mathsf{MA}$; and the prover's advantage for upper bounds is exactly the non-weak-automatability of its proof system. A Python implementation illustrates the gains (e.g. $p \approx 4 \cdot 10^{-46}$ to 0.4% from 10^3 samples) and the soundness of the intervals.

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1 Introduction

Let $C : \{0,1\}^n \to \{0,1\}$ be a Boolean circuit and D a distribution on its inputs. We want $p = \Pr_{x\sim D}[C(x) = 1]$ when p is tiny. Monte Carlo sampling finds it to relative error ε with $\Theta(\log(1/\delta)/(p\varepsilon^2))$ evaluations of C: polynomial in $1/p$, so exponential in the *surprise* $\log_2(1/p)$ of the event. This note asks two questions.

1. *Without help.* For which circuits can a polynomial-time procedure, one that may check some property of the circuit and then sample, estimate p to relative error at a cost polynomial in $\log(1/p)$ instead of $1/p$, while never being wrong (beyond probability δ) on any circuit?

2. *With a prover.* An unbounded prover sends one argument; a polynomial-time verifier checks it and samples. No argument may make the verifier accept a false claim with probability above δ, on any circuit. For which circuits do convincing arguments exist?

These are certified versions of the low-probability estimation problems studied by the Alignment Research Center [WH24, CNX22], where the aim is to estimate the probability of a rare behaviour of a model faster than by sampling, using the model's internals.

1.1 The picture

One idea organises almost everything: **upper bounds need deduction; lower bounds need proposals; sampling pays for whatever surprise is left**.

- A lower bound $p \geq L$ can be certified by exhibiting mass: a proposal distribution concentrated where $C = 1$, with computable likelihood ratios. Clipped importance weights make this sound for any proposal (Proposition 6.6).
- An upper bound below the sampling scale cannot be certified by evaluating C, however cleverly: a verifier that makes T evaluations cannot certify $p < 1/T$ (up to constants), whatever argument it is given (Theorems 3.2 and 6.7). It must read the circuit and *deduce* that $C(x) = 1$ confines x to a region of small, exactly computable measure.
- Combining the two, an *envelope*, a tractable function φ with $\varphi \geq \mathbf{1}_C$ whose mass $Z = \mathbb{E}_D \varphi$ is computable and from which one can sample, yields an importance sampler with bounded weights, hence two-sided intervals, at cost $O(K\varepsilon^{-2} \log(1/\delta))$ where $K = Z/p$ (Theorem 4.3). The envelope explains $\log_2(1/Z)$ bits of the surprise; the remaining $\log_2 K$ bits are paid for by sampling, and this residual cost is unavoidable for anything that treats the unexplained part as a black box (Theorem 6.7).

Karp–Luby–Madras estimation of DNF probabilities [KLM89], exact counting for decision-DNNFs and bounded treewidth, exponential tilting for large deviations [Sie76] and the Chernoff bound are all instances.

1.2 Results

Limits (Section 3). Beating sampling on all circuits is impossible: it would put NP in BPP, and with exponentially hard pseudorandom generators no procedure beats sampling by more than a $p^{-o(1)}$ factor even on explicit small circuits (Proposition 3.1). With a prover, no argument system is complete for upper bounds unless $\mathsf{coNP} \subseteq \mathsf{MA}$ (Theorem 3.4), and a verifier that only evaluates C cannot certify upper bounds below about $1/T$ with T evaluations (Theorem 3.2). Interaction would trivialise the problem (sum-check gives exact p for every circuit), so the one-message model is the interesting one.

Without a prover (Section 5). The procedure DFSB (Decompose–Force–Sample–Branch) recursively splits the event by exact rules (independence of variable-disjoint parts, literals forced by propagation, Shannon expansion) and samples only at leaves that are dense or have tight envelopes. It is sound on every circuit, and complete, with cost $\tilde{O}(m\,K\,d^2\varepsilon^{-2} \log(m/\delta))$ evaluations, on every circuit whose ideal K-explanation tree has m nodes (Theorem 5.2). This covers products and read-once compositions of arbitrary dense sub-circuits, backdoor-style triggers, circuits with small cutsets (exactly, by cutset conditioning), DNFs and DNNFs, and, through sound local rewriting, disguised versions of all of these. Beyond it:

- *Context width* (Theorem 5.6): every circuit compiles into an equivalent DNNF of size $O(|C|\,\mathrm{cw}\,2^{\mathrm{cw}})$, where cw measures how many variables an $\wedge$-gate must be conditioned on

to become decomposable; with the #DNNF FPRAS [MdC26] this gives an FPRAS whenever cw $= O(\log |C|)$, e.g. for ORs of bounded-pathwidth CNFs, whose exact counting is #P-hard.

- *Modules* (Lemma 5.8): functions of independent sub-circuits (thresholds, majorities) are estimated through their quotients, with errors that add.
- *The presumption of independence, certified* (Theorem 5.9): for unate $\wedge$'s of arbitrary sub-circuits, Harris and Janson inequalities pin p to within e^{Δ} of the independence estimate $\prod_i D(A_i)$, using only additive estimates of non-rare quantities; a covariance correction reduces the error to third order, and pairwise data provably cannot do better than $e^{0.29\Delta}$ (Proposition 5.10).
- *Certified telescoping* (Theorem 5.12): annealing estimators become sound when the intermediate samplers are exact Las Vegas samplers (coupling from the past) whose truncation is certified empirically; this handles unate circuits where correlations are strong and extensive.
- Bounded treewidth (exact), thresholds of independent blocks (exponential tilting with $K = O(\sqrt{k})$, Proposition 4.6), and the known FPRAS classes [Moi19, GKM10].

With a prover (Section 6). Proof-carrying importance sampling: the argument is a DAG of deductive steps checked by any Cook–Reckhow proof system (implications, exclusivity, independence) and sampling steps with declared confidence budgets. It is sound on every circuit and every argument (Theorem 6.2), and complete on every circuit that has a short proof of being confined to a tractable region in which it is not rare (Theorem 6.5). Lower bounds need no proof at all. The verifier's cost is $\mathrm{poly}(|\pi|) + 2^{\sigma_u}$, where σ_u is the surprise left unexplained by the argument, and both terms are necessary in the black-box sense: each bit of argument buys at most one bit of surprise (Theorem 6.7). The prover helps strictly: for lower bounds unless $\mathsf{NP} \subseteq \mathsf{BPP}$; for upper bounds exactly to the extent that the verifier's proof system is not weakly automatable (Corollary 6.8); two-sidedly under the hardness of factoring. Such separations cannot be based on NP-hardness alone unless $\mathsf{coNP} \subseteq \mathsf{AM}$.

Experiments (Section 7). A Python implementation (`rareprob`) estimates, e.g., $p \approx 4 \cdot 10^{-46}$ for an OR of eight window-CNFs on 800 variables to 0.4% from 1,024 samples (by context-width compilation), $p \approx 7 \cdot 10^{-25}$ for an 80-bit trigger with $2 \cdot 10^4$ samples, and $p \in [6.2, 7.3] \cdot 10^{-22}$ for a $2 \cdot 10^5$-clause unate CNF deterministically; it certifies SPN-hidden triggers with SAT-solver proofs where all syntactic methods fail; and in 2,450 adversarial and random trials no interval missed p more often than allowed.

1.3 Related work

Approximate counting. Karp–Luby–Madras [KLM89] and the stopping rule of [DKLR00] are the prototypes; for self-reducible classes, approximate counting and approximate sampling are equivalent [JVV86], and Karger's unreliability FPRAS [Kar99] is an early instance of a combinatorial theorem (few near-minimum cuts) serving as the envelope. FPRASs exist for DNNF [MdC26] (and before it nFBDD [MdC25] and NFA [ACJR21]), for CNFs in the local-lemma regime [Moi19, FGYZ21, FHY21, JPV21, HWY22, WY24], and for knapsack [GKM$^+$11, GKM10]; d-DNNF, OBDD, SDD and bounded treewidth admit exact counting [DM02, SS10]. Hashing-based counters [CMV13] use an NP oracle; partial compilation with sampling [LMY23] and

AND/OR importance sampling [GD08] combine decomposition with sampling for variance reduction, without relative-error guarantees for rare events. Certified exact model counting [BNAH23] checks compilation by proofs. *Rare-event simulation* [AG07, Buc04] studies importance sampling with bounded relative error; the soundness requirement here (valid intervals on every input) is the computer-science twist. *Interactive proofs* certify exact counts [LFKN92] and lower bounds [GS86]; the Merlin–Arthur setting connects to proof complexity [CR79, KP98, BPR00, Pud03] and to the class SBP [BGM06]. *Heuristic arguments.* ARC's programme [CNX22, Hil24, WH24] seeks estimators that use a model's internals to beat sampling, judged by "surprise accounting"; this note isolates the part of that programme that can be made sound, and shows where soundness forces deduction. ARC's "matching sampling principle" [NLW+25] asks for estimators that do at least as well as sampling (on average over a distribution of circuits, without soundness), and its "computational no-coincidence principle" [Ney25] conjectures that structure in random reversible circuits always has short explanations checkable by a verifier; our Part 2 is a worst-case, statistically sound relative of the latter, for probabilities instead of properties.

2 Setting

Circuits. A circuit C on inputs $x_1, \dots, x_n$ is a directed acyclic graph whose internal nodes (gates) compute $\wedge$, $\vee$ (any fan-in), $\neg$, $\oplus$ or a threshold function $[\sum_j w_j y_j \geq t]$ of their inputs; one gate is the output. $|C|$ is the number of wires. For a gate g, $\mathrm{var}(g) \subseteq [n]$ is the set of input variables in its cone (its *support*; always the syntactic one). Pushing negations to the inputs gives the negation normal form (NNF) of C: an $\{\wedge, \vee\}$-circuit over literals. For $\{\wedge, \vee, \neg\}$-circuits it has at most two copies of every gate, one per polarity; a $\oplus$-gate of fan-in k expands to $O(k)$ gates per polarity, threshold gates with polynomially bounded integer weights expand by a counter, and other gates stay opaque. An $\wedge$-gate of an NNF is *decomposable* if its children have pairwise disjoint supports, and an $\vee$-gate is *deterministic* if its children are pairwise inconsistent; an NNF all of whose $\wedge$-gates are decomposable is a DNNF, and a d-DNNF if moreover all its $\vee$-gates are deterministic [DM02].

Distributions. D is a product distribution on $\{0,1\}^n$ with $\Pr[x_i = 1] = q_i$. Product structure is what makes "disjoint supports" mean "independent". A distribution given by a sampler circuit $S : \{0,1\}^m \to \{0,1\}^n$ reduces to the uniform case: $\Pr_{x \sim S(U_m)}[C(x) = 1] = \Pr_{u \sim U_m}[(C \circ S)(u) = 1]$. (The reduction can destroy structure; we do not rely on it.) We write

$$p \;=\; p_D(C) \;=\; \Pr_{x \sim D}[C(x) = 1]$$

and think of p as tiny: 2^{-50}, $n^{-\omega(1)}$, or $2^{-\Omega(n)}$. For a set or function E we write $D(E) = \mathbb{E}_D[E]$.

What counts as an answer. Because p can be arbitrarily small, additive error is useless; we ask for relative error. To let a procedure decline, we let it output intervals.

Definition 2.1 (sound interval estimation)**.** An *estimation procedure* gets $(C, D, \varepsilon, \delta)$ and outputs an interval $[L, U] \subseteq [0,1]$ (possibly the trivial $[0,1]$, meaning "no verdict"). It is *sound* if for every circuit C and every D,

$$\Pr\left[p_D(C) \notin [L, U]\right] \leq \delta.$$

It is *complete* on a class $\mathcal{C}$ if for all $C \in \mathcal{C}$, with probability at least $1 - \delta$ the interval is tight: $U \leq (1 + \varepsilon)L$. It is *efficient* if it runs in time $\mathrm{poly}(|C|, 1/\varepsilon, \log(1/\delta))$.

A sound procedure that is complete on $\mathcal{C}$ gives, on $\mathcal{C}$, a $(1 \pm \varepsilon)$-approximation of p with probability $1 - 2\delta$; outside $\mathcal{C}$ its output is still a valid (if loose) interval. Soundness everywhere is what makes the "check a property, then sample" procedures of the problem statement safe to use on circuits we know nothing about.

The baseline. Monte Carlo draws $x^{(1)}, \dots, x^{(N)} \sim D$ and reports the fraction of ones. A relative-error interval needs $N = \Theta\big(\log(1/\delta)/(p\varepsilon^2)\big)$ samples, so its cost is

$$\mathrm{cost}_{\mathrm{MC}} = \Theta\Big(\frac{|C| \log(1/\delta)}{p\,\varepsilon^2}\Big),$$

exponential in the *surprise* $\log_2(1/p)$. The Dagum–Karp–Luby–Ross stopping rule achieves this without knowing p in advance, and their algorithm $\mathcal{AA}$ is optimal up to a constant factor among procedures that only see i.i.d. samples of a $[0,1]$-valued variable [DKLR00] (see [Che07] for a repaired proof of the stopping-rule guarantee); for 0/1 samples of mean $p \ll 1$ the optimal cost is $\Theta(\log(1/\delta)/(p\varepsilon^2))$. Beating sampling therefore requires looking at the circuit.

Definition 2.2 (beating sampling)**.** A procedure beats sampling on a class $\mathcal{C}$ if it is sound, efficient, and complete on $\mathcal{C}$, and $\mathcal{C}$ contains circuits with $p = n^{-\omega(1)}$ (for most of our classes, $p = 2^{-\Omega(n)}$). Its cost is then polynomial in the surprise $\log(1/p)$ rather than in $1/p$.

Prover-assisted estimation. In the second setting an unbounded prover sends one string π (an *argument*) to a randomized polynomial-time verifier V, which then flips coins (for instance to sample) and outputs an interval $[L, U]$.

Definition 2.3 (argument system)**.** V is *sound* if for every circuit C, every D and every string π, $\Pr_{\text{coins of } V}\big[p_D(C) \notin V(C, D, \pi, \varepsilon, \delta)\big] \le \delta$. It is *complete* on $\mathcal{C}$ if for every $C \in \mathcal{C}$ there is a π of size $\mathrm{poly}(|C|)$ such that with probability $1 - \delta$ the output is tight. One-sided variants (lower-bound or upper-bound arguments) are defined in the obvious way.

This is the non-interactive, statistically sound (Merlin–Arthur) model. We will see in Section 3 that interaction (or computational soundness in the random-oracle model) would trivialise the problem, while in this model upper bounds on p cannot be certified for all circuits unless $\mathsf{coNP} \subseteq \mathsf{MA}$.

3 Limits

This section fixes what cannot be done, so that the positive results of Sections 4 to 6 can be read against it.

3.1 Without a prover: structure is necessary

Proposition 3.1 (worst case)**.** *(a) If a sound efficient procedure is complete (for some fixed $\varepsilon > 0$; any finite ε works) on all 3-CNF formulas under the uniform distribution, then* $\mathsf{NP} \subseteq \mathsf{BPP}$.

(b) Let $G_k : \{0,1\}^k \to \{0,1\}^{2k}$ be computable by $\mathrm{poly}(k)$*-size circuits and suppose every circuit of size $T(k)$ distinguishes $G_k(U_k)$ from U_{2k} with advantage less than $1/2$. For $y \in \{0,1\}^{2k}$ let $C_y(x) = [G_k(x) = y]$ on inputs $x \in \{0,1\}^k$, $k \ge 3$. Then $p(C_{G(s)}) \ge 2^{-k}$ for every seed*

s, while $\Pr_y[p(C_y) > 0] \le 2^{-k}$ *for uniform* y*; and any sound procedure with* $\delta \le 1/8$ *that is complete on* $\{C_{G(s)}\}$ *within time* t *(on that family) satisfies* $t \log t \ge T(k)/\mathrm{poly}(k)$*.*

Proof. (a) Answer "satisfiable" iff $L > 0$. If $p = 0$, then $L > 0$ means $p \notin [L, U]$, which has probability at most δ. If $p > 0$ (so $p \ge 2^{-n}$), then $L = 0$ means the output is not tight or does not contain p, which has probability at most 2δ. Run with $\delta = 1/8$. (b) The bounds on p are immediate ($\mathbb{E}_y|G^{-1}(y)| = 2^{k-2k}$). Let A be the procedure, t its running time on the family, and consider the test "$A(C_y)$, stopped after t steps, outputs $L > 0$" (stopping only lowers the firing probability on other inputs). On $y = G(s)$ it fires with probability at least $1 - 2\delta$ (a tight interval containing $p > 0$ has $L > 0$; tightness and containment each fail with probability at most δ). On uniform y, with probability at least $1 - 2^{-k}$ we have $p = 0$, and soundness forces $L \le p = 0$ except with probability δ. The advantage is at least $1 - 3\delta - 2^{-k} \ge 1/2$ for $\delta \le 1/8$, $k \ge 3$. Fixing A's coins to their best value and simulating t steps by a circuit of size $O(t \log t) + \mathrm{poly}(k)$ gives the claim. □

Monte Carlo on $C_{G(s)}$ costs $O(2^k \mathrm{poly}(k))$. So if G is exponentially hard ($T(k) = 2^{(1-o(1))k}$), no sound procedure beats sampling on this family by more than a $2^{o(k)} = p^{-o(1)}$ factor, even though every instance is a small, explicit circuit. Beating sampling is a matter of structure: the classes of Section 5 are where the structure is visible to a polynomial-time algorithm, and those of Section 6 are where it is visible to a polynomial-time verifier given a hint.

3.2 With a prover: upper bounds need deduction

Lower and upper bounds behave very differently. A lower bound $p \ge L$ can be witnessed by *showing* probability mass: a proposal distribution Q concentrated on $C^{-1}(1)$ (Section 6.3). An upper bound must exclude mass that the verifier has not seen. The next theorem makes this precise: a verifier that only evaluates C cannot certify an upper bound much below the reciprocal of its number of evaluations, whatever argument it is given.

Theorem 3.2 (sampling cannot certify upper bounds)**.** *Let* V *be a verifier that receives an argument* π*, has oracle access to* $f : \{0,1\}^n \to \{0,1\}$*, flips coins, makes at most* T *oracle queries (at adaptively chosen points, from any distributions it likes), and outputs a number* U*. Suppose* V *is sound for upper bounds under the uniform distribution: for all* f, π*,* $\Pr[U < p(f)] \le \delta$*. Then for every* f*, every* π *and every* $k \le n$*,*

$$\Pr\left[V^f(\pi) \text{ outputs } U < 2^{-k}\right] \;\le\; \delta + T\,2^{-k}.$$

In particular, for $2^{-(n+1)} \le \eta \le 1$*, to output an upper bound below* η *with probability* $1 - \delta'$ *a black-box verifier needs* $T \ge (1 - \delta - \delta')/(2\eta)$ *queries.*

Proof. Fix f, π, k. For $t \in \{0,1\}^k$ let $H_t = \{x : x_{1..k} = t\}$ and $f_t = f \vee \mathbf{1}_{H_t}$, so $p(f_t) \ge 2^{-k}$. Fix the coins r of V. The run of $V^f(\pi; r)$ queries a fixed sequence of points $z_1, \ldots, z_T$. The run of $V^{f_t}(\pi; r)$ is identical unless some $z_i \in H_t$; for each fixed z_i this happens for exactly one t. Hence for uniform t, $\Pr_{r,t}[\text{the runs differ}] \le T2^{-k}$, and

$$\mathbb{E}_t \Pr_r\left[V^{f_t}(\pi; r) < 2^{-k}\right] \;\ge\; \Pr_r\left[V^f(\pi; r) < 2^{-k}\right] - T2^{-k}.$$

Some t attains at least the average; for it, V outputs $U < 2^{-k} \le p(f_t)$ with that probability, which soundness bounds by δ. For the last claim take $k = \lceil \log_2(1/\eta) \rceil - 1 \le n$, so that $\eta < 2^{-k} \le 2\eta$. □

Remark 3.3. The same argument works for any distribution D with enough collision entropy, planting $H = \{x : Ax = b\}$ for a random $k \times n$ matrix A over $\mathbb{F}_2$ and random b: every fixed point lies in H with probability 2^{-k}, and by pairwise independence $D(H) \geq 2^{-k-1}$ except with probability $4 \cdot 2^k \sum_x D(x)^2$ (for a product distribution, $\sum_x D(x)^2 = \prod_i (q_i^2 + (1-q_i)^2)$). The bound becomes $\Pr[V^f(\pi) \text{ outputs } U < 2^{-k-1}] \leq \delta + T2^{-k} + 4 \cdot 2^k \sum_x D(x)^2$. It also extends to verifiers that see a circuit for f and its size but use the circuit only through evaluations, provided the planted gadget is padded to a topology independent of (A, b) (each row as $\bigoplus_j (x_j \wedge a_{ij}) \oplus b_i$ with constant leaves a_{ij}, b_i): then the bound holds for *some* circuit for f of size $|C| + O(nk)$, namely $C \vee (0 \wedge G_{A_0,b_0})$. It cannot hold for every circuit computing f: a verifier that sees $|C| = 3$ knows $p \in \{0\} \cup [1/8, 1]$.

The verifiers of Section 6 escape Theorem 3.2 by reading the circuit: they check a proof that $C(x) = 1$ forces x into a region of small, exactly computable measure. In general such proofs cannot always exist:

Theorem 3.4 (no complete upper-bound arguments)**.** *Suppose a sound argument system (soundness error $\delta \leq 1/3$) is complete for upper bounds on all 3-CNF formulas with $p = 0$ under the uniform distribution, i.e. for each such C some π of polynomial size makes V output $U < 2^{-n}$ with probability at least $2/3$. Then* coNP $\subseteq$ MA*, and the polynomial hierarchy collapses to its second level.*

Proof. Merlin sends π; Arthur runs V and accepts iff $U < 2^{-n}$. If C is unsatisfiable he accepts with probability $\geq 2/3$. If C is satisfiable then $p \geq 2^{-n}$, and accepting means V output $U < p$, which happens with probability at most $1/3$. So 3-UNSAT$\in$ MA $\subseteq$ AM, and coNP $\subseteq$ AM collapses PH to Σ_2^p [BHZ87] (in fact coNP $\subseteq$ MA gives PH $=$ MA). □

The lower-bound side is weaker but not free: a system complete for lower bounds within a constant factor on all circuits exists iff prSBP $\subseteq$ prMA for the class SBP of [BGM06] (amplify the SBP gap by conjunctions of copies; conversely send the scale with an MA proof), and in particular it would give SBP $\subseteq$ MA. Since MA $\subseteq$ SBP $\subseteq$ AM, with an oracle relative to which SBP $\not\subseteq \Sigma_2^p$, we should not expect proposal-based lower bounds for every circuit either.

3.3 Interaction trivialises the problem

The restriction to one message is essential:

Proposition 3.5. *(a) (Sum-check [LFKN92].) For every circuit C and product D with dyadic marginals (total denominator bits B), after threshold gates are compiled into $\{\wedge, \vee, \neg, \oplus\}$-subcircuits C', the exact value of p has an interactive proof over a prime field with $|\mathbb{F}| > 2^{B+1}$, with a polynomial-time verifier, $O(n + |C'|)$ rounds and soundness error $O((n + |C'|)\, d/|\mathbb{F}|)$, where $d \leq |C'| + 2$ bounds the individual degrees of the arithmetised Tseitin summand.*

(b) (Set lower bounds [GS86].) For uniform D (or dyadic product D, via the sampler reduction), two-round public-coin protocols, repeated in parallel, certify $p \geq L/c$ for a constant c for every circuit.

Applying the Fiat–Shamir transform to (a) gives, in the random-oracle model (or heuristically with a fixed hash), a one-message argument that is only computationally sound, which is not the model asked for. So the interesting question is the Merlin–Arthur one, where by Theorems 3.2

and 3.4 an argument certifying upper bounds below the sampling scale must contain *deductive* content about the circuit, and, unless coNP $\subseteq$ MA, can be complete for upper bounds only on a restricted class.

4 Envelopes: the common mechanism

Most of our upper-bound-capable methods, with and without a prover, are instances of one estimator (the exceptions are the deterministic correlation-inequality bounds of Sections 4.5 and 5.5, the counter-guided sampler of Theorem 6.4, which is a variant of it, and the product-of-ratios estimators of Section 5.6). It turns a *deductive* fact ("$C(x) = 1$ implies $\varphi(x) \geq 1$") and an *exact* computation ($\mathbb{E}_D\varphi$) into an importance sampler whose cost depends only on how loose the deduction is.

4.1 Envelope circuits

Definition 4.1 (envelope)**.** An *envelope circuit* Φ is an arithmetic circuit whose leaves are literal indicators $[x_i = b]$, nonnegative rational constants, and (Section 4.3) *boxes* $[B(x) = 1]$ for circuits B; whose gates are SUM and PROD; and whose PROD gates are *decomposable*: their children have pairwise disjoint supports, where the support $\mathrm{var}(v)$ of a node is the set of variables occurring at the leaves of its cone ($\mathrm{var}(B)$ for a box). We also require that every PROD-child with empty support be a constant leaf (fold constant subcircuits first). Φ computes $\varphi : \{0,1\}^n \to \mathbb{R}_{\geq 0}$. It *dominates* C if $\varphi(x) \geq 1$ whenever $C(x) = 1$. Its *mass* is $Z = \mathbb{E}_D\varphi$; if $p > 0$, its *looseness on* C is $K = Z/p$, and domination gives $Z \geq p$, i.e. $K \geq 1$.

A *derivation* of Φ is a proof tree of its tree unfolding: keep, top-down, one child of every SUM occurrence and all children of every PROD occurrence. By decomposability a derivation visits each node of nonempty support at most once. With literal leaves and unit constants, $\varphi(x)$ is the number of derivations all of whose leaves are satisfied by x; with general constants it is the corresponding weighted count. A DNF $\bigvee_j T_j$ with consistent, duplicate-free terms gives the envelope $\sum_j \prod_{\ell \in T_j} [\ell]$, and a DNNF gives the envelope obtained by reading $\vee$ as SUM and $\wedge$ as PROD.

Lemma 4.2 (exact mass, exact sampling)**.** *Let Φ have literal and constant leaves only, D be a product distribution with rational marginals, and $Z > 0$. Then $Z = \mathbb{E}_D\varphi$ is computed by one bottom-up pass (SUM $\mapsto$ sum of children's masses, PROD $\mapsto$ product) with $O(|\Phi|)$ operations on rationals of* $\mathrm{poly}(|\Phi|, b)$ *bits (b the bit length of the marginals), and the following top-down procedure samples exactly from $Q_\varphi(x) = D(x)\varphi(x)/Z$: at a SUM gate choose a child with probability proportional to its mass; at a PROD gate visit all children; at a literal leaf set the literal; finally draw every unset variable from D.*

Proof. By decomposability and independence, $\mathbb{E}_D \prod_c \varphi_c = \prod_c \mathbb{E}_D\varphi_c$; linearity handles SUM. The procedure selects a derivation $\mathcal{T}$ with probability $w(\mathcal{T})/Z$, where $w(\mathcal{T})$ is the product of the literals' probabilities and the constants in $\mathcal{T}$ (the choice probabilities telescope), and then draws $x \sim D$ conditioned on all literals of $\mathcal{T}$, whose density is $D(x)\mathbf{1}[x \models \mathcal{T}]\, c(\mathcal{T})/w(\mathcal{T})$ with $c(\mathcal{T})$ the product of its constants. Summing over derivations, $\Pr[x] = \sum_{\mathcal{T}} c(\mathcal{T})D(x)\mathbf{1}[x \models \mathcal{T}]/Z = D(x)\varphi(x)/Z$. (A literal is set at most once per derivation: two literals on the same variable in one derivation would lie below different children of a decomposable PROD.) Bit lengths: the mass of a node is a sum of at most $2^{|\Phi|}$ products of at most $|\Phi|$ constants and marginals. □

In floating point, masses must be kept in the log domain and the final interval widened by the accumulated rounding error; our implementation computes Z exactly with rational arithmetic and rounds the output interval outwards.

4.2 The envelope estimator

For N i.i.d. samples in $[0,1]$ with empirical mean $\hat{\mu}$, the *Hoeffding–KL interval* at level δ is $\{m \in [0,1] : N\,\mathrm{KL}(\hat{\mu}\|m) \le \ln(2/\delta)\}$ with KL the Bernoulli relative entropy; it contains the true mean with probability at least $1-\delta$ [Hoe63, Thm. 1].

Theorem 4.3 (envelope estimator)**.** *Let $\varphi : \{0,1\}^n \to \mathbb{R}_{\ge 0}$ dominate C, with exact mass Z and an exact sampler for $Q_\varphi = D\varphi/Z$ (for instance an envelope circuit with Lemma 4.2). If $Z = 0$, output $[0,0]$ (valid under domination). Otherwise, for $X \sim Q_\varphi$ let $W = C(X)/\varphi(X)$. Then*

(i) $W \in [0,1]$ and $\mathbb{E}W = p/Z = 1/K$;

(ii) from N independent copies, the Hoeffding–KL interval $[\ell, u]$ for $\mathbb{E}W$ at level δ gives an interval $[Z\ell, \min(Zu, 1)]$ containing p with probability at least $1-\delta$, for every circuit C *as long as φ dominates it;*

(iii) if φ does not dominate C: for integer-valued φ (envelope circuits with literal leaves and integer constants) the same procedure gives a valid interval for $\Pr_D[C(x) = 1 \wedge \varphi(x) > 0] \le p$; for general φ, with W clipped at 1, $Z\ell$ is still a valid lower bound on p;

(iv) for $0 < \varepsilon \le 1$, run sequentially from $N_0 = \lceil c \log(2/\delta_1)/\varepsilon^2 \rceil$ samples, doubling N and using level $\delta_j = 6\delta/(\pi^2 j^2)$ at stage j: all intervals are valid with probability $1-\delta$, and if $p > 0$ the procedure reaches $u \le (1+\varepsilon)\ell$ after $N = O(K(\log(1/\delta) + \log\log K)/\varepsilon^2)$ samples with probability $1-\delta$ (so both hold with probability $1-2\delta$). With a sample cap it always halts, returning its current, still valid, interval. Each sample costs $O(|C| + |\Phi|)$ arithmetic operations with the sampler of Lemma 4.2.

Proof. (i) If $Q_\varphi(x) > 0$ then $\varphi(x) > 0$; if moreover $C(x) = 1$ then $\varphi(x) \ge 1$, so $W \le 1$. $\mathbb{E}W = \sum_x D(x)\varphi(x)C(x)/(Z\varphi(x)) = p/Z$. (ii) The KL interval is valid for i.i.d. $[0,1]$ variables, and $p \le 1$. (iii) For integer-valued φ, $\varphi(x) > 0$ implies $\varphi(x) \ge 1$, so $W \in [0,1]$ and $\mathbb{E}W = \Pr[C = 1, \varphi > 0]/Z$. For real-valued φ, clipping only lowers the mean, so $\mathbb{E}\min(W,1) \le \Pr[C = 1, \varphi > 0]/Z \le p/Z$. (iv) Let $\mu = 1/K$. By the multiplicative Chernoff bound, once $N \ge c\log(1/\delta_j)/(\mu\varepsilon^2)$ the empirical mean $\hat{\mu}$ is within a factor $1 \pm \varepsilon/8$ of μ with probability $1-\delta_j$. Since $\mathrm{KL}(a\|b) \ge (a-b)^2/(2\max(a,b))$, every m in the KL interval with $m > \hat{\mu}$ satisfies $(m-\hat{\mu})^2 \le am$ with $a = 2\log(2/\delta_j)/N$, hence $m - \hat{\mu} \le a + \sqrt{a\hat{\mu}}$, and similarly below; for a suitable constant c both are at most $(\varepsilon/4)\hat{\mu}$, so $u/\ell \le (1+\varepsilon/4)/(1-\varepsilon/4) \le 1+\varepsilon$. With the stated N_0, the first stage at which this holds has $j = O(\log K)$, whence the $\log\log K$ term; the union bound over stages makes all intervals simultaneously valid. □

So the cost of an envelope is governed by one number. Monte Carlo is the trivial envelope $\varphi \equiv 1$, with $K = 1/p$; exact computation is $\varphi = C$ (when C is itself a d-DNNF), with $K = 1$. In between:

Lemma 4.4 (what K measures)**.** *For $p > 0$, $K = \mathbb{E}_D[\varphi \mid C = 1] + \mathbb{E}_D[\varphi\,\mathbf{1}_{C=0}]/p$. For an envelope with literal leaves whose positivity set $\{\varphi > 0\}$ equals $C^{-1}(1)$ (e.g. a DNNF for C), K is the expected number of satisfied derivations of $x \sim D \mid C = 1$, its* expected ambiguity*. For a DNF with m terms, $K \le m$.*

The first term is overcounting (union bounds); the second is mass that the deduction failed to exclude. K also bounds the relative variance $\rho = \frac{Z}{p}\mathbb{E}_D[1/\varphi \mid C = 1] - 1 \le K - 1$ of ZW. The KL procedure of Theorem 4.3(iv) ignores the variance and pays $\Theta(K)$ samples whatever ρ is; a variance-adaptive interval (empirical Bernstein, or DKLR's algorithm $\mathcal{AA}$) pays $O((\rho/\varepsilon^2 + K/\varepsilon)\log(1/\delta))$, so a low dispersion of the derivation count helps only in the ε^{-2} term. A d-DNNF has $\varphi = \mathbf{1}_C$, $K = 1$ and $\rho = 0$. For $E = \bigwedge_{i \le m}(x_i \vee y_i)$ under the uniform distribution, used as its own envelope, $K = (4/3)^m$, and the relative variance is at least $(10/9)^m - 1$ under every reweighting of its $\vee$-edges (with equality for unit weights); both are exponential although $D(E) = (3/4)^m$ is known exactly. Theorem 6.4 removes this overcounting altogether: its looseness is $D(E)/p$ rather than $\mathbb{E}_D[\varphi]/p$.

In the spirit of surprise accounting [Hil24]: of the $\log_2(1/p)$ bits of surprise in the event $C(x) = 1$, an envelope with $Z \le 1$ explains $\log_2(1/Z)$ bits by deduction and exact computation (once domination is certified, syntactically or by a proof), and leaves $\log_2 K$ bits to sampling, at a cost of about K samples (up to $\varepsilon^{-2}\log(1/\delta)$ factors). If $Z \ge 1$ the envelope is worse than Monte Carlo and is replaced by $\varphi \equiv 1$. Monte Carlo explains nothing; exact computation explains everything, with $|\Phi|$ possibly exponential in $|C|$. Surprise accounting also charges the explanation's own description length: zero for envelopes computed from C, and $|\pi|$ for prover-supplied ones (Theorem 6.7).

4.3 Dense leaves

Products of independent events that are individually not rare are the most common source of rarity, and they need not have small envelopes: an $\wedge$ of k independent random sub-circuits B_j, each true with probability $1/2$, has $p = 2^{-k}$ but its literal envelope may lose a constant factor per block. Boxes repair this: a box is a sub-function used as its own envelope, its mass estimated by Monte Carlo and its conditional distribution sampled by rejection. Estimation errors in the masses bias the proposal, but the bias on each derivation is a product of per-box ratios that we can bound exactly.

Theorem 4.5 (envelopes with dense leaves)**.** *Let Φ dominate C, with literal leaves and box leaves $B_1, \dots, B_r$ (PROD gates decomposable with respect to supports), where $m_j = D(B_j) \ge 1/R$ and every derivation contains at most L box leaves; let $0 < \varepsilon \le 1$. Estimate each m_j by Monte Carlo to an interval $[\ell_j, u_j]$ at level $\delta/(2r)$ with point value $\hat{m}_j$; compute masses $\hat{Z}$ and sample $\hat{Q}$ as in Lemma 4.2 with $\hat{m}_j$ in place of m_j, filling each box by rejection sampling from D. For $\rho(\mathcal{T}) = \prod_{B_j \in \mathcal{T}} \hat{m}_j/m_j$ let $\rho_{\min}, \rho_{\max}$ be its extreme values over derivations when each m_j ranges over $[\ell_j, u_j]$ (computed by a min-/max-product pass over Φ). Then, with $[\ell, u]$ a valid KL interval for $\mathbb{E}_{\hat{Q}}W$, $W = C/\varphi$, at total level $\delta/2$ (fixed or sequential),*

$$p \in \left[\hat{Z}\ell/\rho_{\max},\ \hat{Z}u/\rho_{\min}\right]$$

with probability at least $1-\delta$. If each box interval satisfies $u_j \le (1+\varepsilon/(8L))\ell_j$ and the importance-sampling stage is run to ratio $1+\varepsilon/4$, the output is tight with probability $1-2\delta$, after $O(K(\log(1/\delta)+\log\log K)/\varepsilon^2)$ samples of $\hat{Q}$, each costing $O(|C| + |\Phi| + \sum_j |B_j| + LR\max_j |B_j|)$ in expectation; estimating the boxes costs $O(rL^2R\log(r/\delta)/\varepsilon^2)$ evaluations.

Proof. The sampler picks derivation $\mathcal{T}$ with probability $\hat{w}(\mathcal{T})/\hat{Z}$ where $\hat{w}$ uses the $\hat{m}_j$, and then draws x exactly from D conditioned on $\mathcal{T}$'s leaves (rejection sampling is exact; decomposability makes the leaves' conditionings independent), whose density is $D(x)\mathbf{1}[x \models \mathcal{T}]/w(\mathcal{T})$ with the true

masses. Hence $\hat{Q}(x) = \frac{D(x)}{\hat{Z}} \sum_{\mathcal{T} \ni x} \rho(\mathcal{T})$, and

$$\hat{Z}\, \mathbb{E}_{\hat{Q}} W = \sum_x D(x)C(x) \cdot \frac{1}{\varphi(x)} \sum_{\mathcal{T} \ni x} \rho(\mathcal{T}) \;\in\; [\rho_{\min}, \rho_{\max}] \cdot p,$$

because $\varphi(x)$ is the number of derivations satisfied by x, so the middle factor is an average of $\rho(\mathcal{T})$. On the event (probability $\geq 1 - \delta/2$) that all $m_j \in [\ell_j, u_j]$, $\rho_{\min}$ and $\rho_{\max}$ bound this average; conditionally on the box estimates the KL interval fails with probability at most $\delta/2$. For tightness: $\rho_{\max}/\rho_{\min} \leq (1 + \varepsilon/(8L))^{2L} \leq e^{\varepsilon/4}$ and $(1 + \varepsilon/4)e^{\varepsilon/4} \leq 1 + \varepsilon$ for $\varepsilon \leq 1$. Since $\hat{Z} = \sum_{\mathcal{T}} \rho(\mathcal{T}) w(\mathcal{T}) \leq \rho_{\max} Z$, the mean of W under $\hat{Q}$ is at least $\rho_{\min} p/(\rho_{\max} Z) \geq e^{-\varepsilon/4}/K$, so Theorem 4.3(iv) (which only needs i.i.d. $[0,1]$ samples of that mean) gives the sample count. Computing $\varphi(X)$ evaluates every box, hence the $\sum_j |B_j|$ term. □

With boxes, the product of k independent dense blocks has $K = 1$ and costs $O(k^3 R \log(k/\delta)/\varepsilon^2)$ evaluations, against $2^{\Theta(k)}$ for Monte Carlo; a DNF whose terms are products of dense blocks has an envelope (a SUM of PRODs of boxes) whose positivity set is $C^{-1}(1)$ and with $\varphi \leq m$, so $K \leq m$ by Lemma 4.4.

4.4 Soft envelopes: Chernoff bounds as certificates

Envelopes need not be integer-valued. If $C(x) = [S(x) \geq t]$ with $S = \sum_{j=1}^{k} g_j(x_{B_j})$ a sum of $\{0,1\}$-valued sub-circuits on disjoint blocks, then for $\theta \geq 0$

$$\varphi_\theta(x) = e^{-\theta t} \prod_{j=1}^{k} ([g_j = 0] + e^{\theta}[g_j = 1]) = e^{\theta(S(x)-t)} \;\geq\; \mathbf{1}[S(x) \geq t], \qquad Z_\theta = e^{-\theta t} \prod_{j=1}^{k} (1 - p_j + p_j e^{\theta}),$$

where $p_j = D(g_j = 1)$. Z_θ is the Chernoff bound, a deductively valid upper bound, and Q_{φ_θ} is the exponentially tilted product distribution. To sample it block by block we assume that the p_j are known exactly and D conditioned on $g_j = b$ can be sampled, e.g. blocks small enough to enumerate (as in our implementation), or boxes with $\min(p_j, 1 - p_j) \geq 1/R$, in which case Theorem 4.5 applies with $L = k$ (with weighted averages of ρ). This is the classical efficient importance sampler for large deviations of sums [Sie76, Buc04]; here is a clean uniform bound on its looseness.

Proposition 4.6 (tilting is polynomially tight)**.** *Let the weights be 1, $\mathbb{E}_D S \leq t < k_+$, where $k_+ = |\{j : p_j > 0\}|$ and t is an integer. There is $\theta \geq 0$ with $\mathbb{E}_{Q_\theta} S = t$, and for it $K = Z_\theta/p \leq \frac{4}{3}(2\sqrt{k} + 1)$. The same bound holds for any $\theta \geq 0$ with $|\mathbb{E}_{Q_\theta} S - t| < 1/(k+2)$, so θ may be found by bisection and e^θ taken rational.*

Proof. $\theta \mapsto \mathbb{E}_{Q_\theta} S = \sum_j p_j e^\theta/(1 - p_j + p_j e^\theta)$ is continuous and nondecreasing, at most t at $\theta = 0$ and tends to $k_+ > t$, so θ exists. Under Q_θ the g_j are independent Bernoulli(p_j^θ), so S is Poisson-binomial with mean μ_θ. By Darroch's rule [Dar64] the mode is t when $\mu_\theta = t$, and also when $|\mu_\theta - t| < 1/(k+2)$ (the rule makes k' the unique mode for $k' \leq \mu < k' + 1/(k'+2)$ and for $k' - 1/(n - k' + 2) < \mu \leq k'$). By Chebyshev, $Q_\theta(|S - \mu_\theta| \leq 2\sigma) \geq 3/4$ with $\sigma^2 = \mathrm{Var}_{Q_\theta} S \leq k/4$, and this window contains at most $4\sigma + 1$ integers, so, S being unimodal (log-concave), $Q_\theta(S = t) \geq \frac{3}{4(4\sigma+1)}$. Finally $p = Z_\theta\, \mathbb{E}_{Q_\theta}[e^{-\theta(S-t)} \mathbf{1}_{S \geq t}] \geq Z_\theta\, Q_\theta(S = t)$. □

Unit weights matter. For integer weights under a product distribution the best tilt can be loose by a factor $p^{-\Theta(1)}$: with $S = (k+1)y + \sum_{i \leq k} x_i$, $q_y = \eta$, $q_x = 1/2$ and $t = k+1$, so $p = \eta$, the

optimal Chernoff envelope has $K = 2^{25.7}$ at $k = 20$, $p = 2^{-40}$. When the p_j are known exactly, the unit-weight case is also computable exactly by dynamic programming over S (pseudo-polynomial in general weights), and real weights have a deterministic FPTAS under product distributions via read-once branching programs [GKM10] (see also [GKM+11]). The point of Proposition 4.6 is that the Chernoff bound, a one-line deduction from the syntax "C is a threshold of independent blocks", explains all but $O(\log k)$ bits of the surprise, and sampling pays for the rest.

4.5 Certified presumption of independence

The *presumption of independence* [CNX22] estimates $\Pr[\bigwedge_i A_i]$ by $\prod_i \Pr[A_i]$. For monotone events it can be certified: correlation inequalities bound the error in terms of the probability carried by dependent pairs, which can be computed or estimated without ever conditioning on the rare event.

Theorem 4.7 (Harris–Janson sandwich)**.** *Let $C = \bigwedge_{i=1}^{m} \mathrm{cl}_i$ be a CNF in which every variable occurs with only one sign, $B_i = [\mathrm{cl}_i = 0]$ (a subcube), D a product distribution, $b_i = D(B_i)$, $M = \prod_i (1 - b_i)$, $\varepsilon_0 = \max_i b_i < 1$, and for an ordering of the clauses $d_i = \sum_{j<i,\, \mathrm{var}(\mathrm{cl}_j) \cap \mathrm{var}(\mathrm{cl}_i) \neq \emptyset} D(B_i \wedge B_j)$, $\Delta = 2 \sum_i d_i$ (the sum over ordered dependent pairs). Then*

$$M \;\le\; p \;\le\; \prod_i (1 - b_i + d_i) \;\le\; M \exp\Big(\sum_i \frac{d_i}{1 - b_i}\Big) \;\le\; M \exp\Big(\frac{\Delta}{2(1 - \varepsilon_0)}\Big),$$

and all quantities are computable with $O(\sum_i |\mathrm{cl}_i|^2 \cdot d)$ arithmetic operations, d the maximum number of clauses sharing a variable. (If some $b_i = 1$ then $p = M = 0$.)

Proof. Flip the variables that occur positively; then every B_i is a principal up-set $\{y : y_{S_i} = \mathbf{1}\}$ under a product measure. The lower bound is the Harris–Kleitman inequality (the events $\neg B_i$ are decreasing, hence positively correlated) [Har60]. The upper bounds are Janson's inequality [Jan90] in the product form of the Boppana–Spencer proof (Theorem 5.9) and in the form of [AS16, Thm. 8.1.1]. $D(B_i)$ and $D(B_i \wedge B_j)$ are products of marginals over S_i and $S_i \cup S_j$. □

So for unate CNFs with $\varepsilon_0 \le 1/2$ whose dependent pairs carry total probability $\Delta \le \varepsilon$, the geometric mean of the two ends has relative error at most $e^{\Delta/2} - 1$, however small p is. (Without a bound on ε_0 the last form can be arbitrarily loose; the product form, with clauses ordered by decreasing b_i, stays tight.) Example: $m = 6 \cdot 10^4$ random clauses of width 12 over $2 \cdot 10^5$ variables give the certified interval $p \in [4.34, 5.06] \cdot 10^{-7}$ (ratio 1.17) in about 30 seconds of pure Python (Section 7). The version for unate $\wedge$'s of arbitrary monotone sub-circuits, with b_i and Δ estimated by sampling, is Theorem 5.9.

5 Without a prover: sound procedures and their classes

A procedure without a prover must find the structure itself. Ours look for exactly the structure that Section 4 can exploit: independence (variable-disjoint sub-circuits), forcing (literals implied by $C(x) = 1$), small cutsets, dense residuals, and a few algebraic and analytic patterns. All steps are either exact identities under product distributions or Monte Carlo estimates with valid intervals, so the procedures are sound on every circuit; which circuits they are *complete* on is characterised by the size of a canonical "explanation tree".

5.1 Decompose–Force–Sample–Branch

Work with the NNF of C, flattened and with constants propagated; restricting a node by a partial assignment ρ simplifies it again. For a node h:

(T) h is a constant or a literal: exact value.

(F) Unit propagation on $\mathrm{Tseitin}(h) \wedge [h = 1]$ derives a nonempty set of input literals F (at the root, also failed-literal probing and Gaussian elimination on conjuncts that are affine over $\mathbb{F}_2$): $D(h) = D(F) \cdot D(h|_F)$, since $h \Rightarrow F$ and $h|_F$ does not depend on the variables of F.

(D) The children of h's top gate fall into at least two variable-disjoint groups: $D(h) = \prod_j D(h_j)$ for $\wedge$ and $D(h) = 1 - \prod_j (1 - D(h_j))$ for $\vee$.

(S) Dense test: a pilot of $N_0 = 64K\lceil \ln(2m/\delta) \rceil$ samples finds at least $N_0/(4K)$ hits; then h is a Monte Carlo leaf.

(E) Envelope test: a syntactic envelope Φ_h (Section 5.2) passes the analogous pilot for $\mathbb{E}_{Q_\Phi}[h/\varphi]$ (at least $N_0/(4K)$ total weight in N_0 samples); then h is an envelope leaf.

(B) Otherwise branch on the variable shared by the most children: $D(h) = (1 - q_x)D(h|_{x=0}) + q_x D(h|_{x=1})$.

Phase 1 builds the tree with a node budget m; nodes beyond the budget become "loose" leaves. Phase 2 estimates every leaf with fresh samples, to relative accuracy $\varepsilon/(3d)$ at confidence δ/ℓ, where ℓ is the number of leaves and d the largest number of leaves on one derivation (all children at (D)-$\wedge$, one child elsewhere), and combines the leaf intervals through the exact node operations, all of which are monotone.

Definition 5.1 (ideal explanation tree)**.** $T^*_K(C)$ is the tree built by the same rules with the pilot test (S) replaced by the exact predicate $D(h) \geq 1/K$ and without (E). $m_K(C) = |T^*_K(C)|$.

Theorem 5.2 (DFSB)**.** *For every circuit C and product distribution D, the output interval contains p with probability at least $1 - \delta$. If $m_K(C) \leq m$, then with probability at least $1 - 2\delta$ the output is tight ($U \leq (1 + \varepsilon)L$), and the number of circuit evaluations is*

$$O\big(mK \log(m/\delta)\big) + \tilde{O}\Big(\ell \cdot K\, d^2\, \varepsilon^{-2} \log(\ell/\delta)\Big), \qquad \ell \leq m,\ d \leq n,$$

each costing $O(|C|)$; the non-sampling work is $O(m \cdot |C|)$.

Proof. Soundness. (T), (F), (D), (B) are exact identities under a product distribution (independence of variable-disjoint sub-circuits, the law of total probability, and $h \Rightarrow F$). Phase 2 runs on a fixed tree with fresh samples, so a union bound over its ℓ leaves gives probability $1 - \delta$ that every leaf interval is valid; interval arithmetic through monotone operations then gives a valid root interval. Phase 1 only shapes the tree.

Completeness. Let X be a pilot's hit count and $\mu = \mathbb{E}X$. If $D(h) \geq 1/K$ then $\mu \geq N_0/K$ and $\Pr[X < N_0/(4K)] \leq e^{-(3/4)^2\mu/2} \leq (\delta/2m)^{18}$. If $D(h) < 1/(16K)$ then $\mu < N_0/(16K)$ and, by the Chernoff bound $\Pr[X \geq t] \leq e^{-\mu}(e\mu/t)^t$, $\Pr[X \geq N_0/(4K)] \leq (e/4)^{N_0/(4K)} \leq (\delta/2m)^6$. The same two bounds hold for the envelope pilot, whose summands lie in $[0, 1]$. So with probability $1 - \delta$ every pilot behaves: it accepts every node of density $\geq 1/K$ and no node of density $< 1/(16K)$. On that event the actual tree is a pruning of $T^*_K(C)$ (where the ideal tree stops, the actual one

stops; where it continues, the actual one continues identically or stops early at a leaf of density $\geq 1/(16K)$), so it has at most m nodes and every leaf is $16K$-dense (for envelope leaves: has looseness at most $16K$). Each leaf then needs $\tilde{O}(Kd^2\varepsilon^{-2}\log(\ell/\delta))$ samples for relative accuracy $\varepsilon/(3d)$ (Theorem 4.3(iv)), and $(1+\varepsilon/(3d))^d \leq e^{\varepsilon/3}$, so the root ratio is at most $1+\varepsilon$: products multiply leaf ratios; positive combinations with exact weights do not increase them; and for $f(x) = 1 - \prod_j (1-x_j)$, which is monotone and concave along rays with $f(0) = 0$, $x' \leq bx$ implies $f(x') \leq bf(x)$ and $x' \geq ax$ implies $f(x') \geq af(x)$, so (D)-$\vee$ nodes do not increase ratios either. □

The class on which DFSB beats sampling is therefore $\{C : m_K(C) \leq \text{poly}(|C|)$ for some $K \leq \text{poly}(|C|)\}$. It contains:

Corollary 5.3. *Let $s = |C|$.*

(a) Component-dense circuits. *If every leaf of the syntactic component tree of C (built by (F) and (D) alone) has probability at least $1/K$, then $m_K(C) = O(s)$. This includes read-once formulas (exactly) and read-once compositions $F(\lambda_1, \ldots, \lambda_t)$ of arbitrary K-dense circuits on disjoint variable sets; e.g. an $\wedge$ of t blocks of probability $1/2$ has $p = 2^{-t}$ and costs $O(t^3\varepsilon^{-2}\log(t/\delta))$ evaluations.*

(b) Backbones. *If (F) derives a set F with $D(C \mid F) \geq 1/K$ (for instance trigger circuits $R \wedge [x_A = a]$, however the test is written, as long as propagation sees it), then $m_K(C) = O(1)$ and the cost is $O(K\varepsilon^{-2}\log(1/\delta))$ against $2^{|A|}K\varepsilon^{-2}\log(1/\delta)$ for Monte Carlo.*

(c) Cutsets. *Let $\text{Conf}(C)$ be the set of input variables that lie in the supports of two different children of some gate of the NNF. If $|\text{Conf}(C)| = r$ then $m_K(C) = O(s\,3^r)$ for every K, and with $K = 1$ DFSB is exact cutset conditioning [Pea88]: no sampling at all.*

(d) Global variables. *If fixing some set Z of r variables makes C satisfy (a), the variant that first enumerates Z uses $O(2^r s)$ nodes.*

Proof. (a) Every node that reaches (S) is a force-free, non-decomposable leaf of the component tree, hence accepted; forcing never lowers a part's probability ($D(\lambda \mid \ell) = D(\lambda)/D(\ell) \geq D(\lambda)$ for $\lambda \Rightarrow \ell$); tree nodes map injectively to gates, child groups, or force wrappers. (c) Branching happens only at a non-decomposable node, on a variable shared by two children, which lies in $\text{Conf}(C)$ (restriction only shrinks supports); two distinct tree nodes cannot carry the same NNF node and the same partial assignment to $\text{Conf}(C)$, and once all of $\text{Conf}(C) \cap \text{var}(h)$ is assigned every gate is decomposable, so with $K = 1$ the tree ends in literals. There are $O(s)$ NNF nodes and 3^r partial assignments. (b), (d) are immediate. □

5.2 Envelope leaves and sound rewriting rules

Nodes that are rare but not decomposable get an envelope from the following rules, each of which produces $\varphi \geq \mathbf{1}_h$ pointwise, so any combination is sound (Theorem 4.3):

(R1) literal $\mapsto$ literal; $\vee(c_1, \ldots, c_k) \mapsto \sum_j \varphi_{c_j}$ (union bound);

(R2) $\wedge$ with forced literals $F \mapsto \prod_{\ell \in F}[\ell] \cdot \varphi_{h|_F}$;

(R3) $\wedge$ with variable-disjoint children $\mapsto \prod_j \varphi_{c_j}$;

(R4) $\wedge$ with a shared variable u (at most r times on a path) $\mapsto \sum_b [u = b] \cdot \varphi_{h|_{u=b}}$ (Shannon expansion);

(R5) $\wedge$ otherwise $\mapsto$ product over a greedy variable-disjoint subset of conjuncts, most informative first (dropping conjuncts);

(R6) anything $\mapsto 1$, used whenever the rules give mass > 1;

(R7) a dense node $\mapsto$ a box (Theorem 4.5);

(R8) *small-cone canonicalisation*: a gate whose support has at most $c \log |C|$ variables is replaced, before anything else, by its exact reduced OBDD (computed by tabulation). This makes every class above closed under local rewriting: e.g. literals disguised as $(x \oplus r) \oplus r$, or terms built from redundant small gadgets, are recognised exactly.

Proposition 5.4 (DNNF envelopes)**.** *If the NNF of C is a DNNF (after (R8)), rules (R1)–(R3) produce its own arithmetic circuit, with $K = \mathbb{E}_D[\varphi \mid C = 1]$, its expected ambiguity; for a DNF with m terms, $K \le m$ (Karp–Luby–Madras). For DNNFs of large ambiguity dispersion, the FPRAS of [MdC26] with the sampler of Theorem 6.4 gives cost linear in $D(E)/p = 1$; all DNNFs therefore admit polynomial-time relative-error estimation, and so do all circuits that (R1)–(R8) turn into a DNNF envelope E with $D(E) \le Kp$.*

5.3 Context width: compiling any circuit into a DNNF

DFSB branches on shared variables but treats overlapping $\vee$'s by envelopes. With the #DNNF FPRAS available, non-deterministic $\vee$'s cost nothing, and the only obstacle is $\wedge$-gates whose children share variables. The following parameter measures exactly that, and generalises Darwiche's dtree conditioning [Dar01] from CNFs to arbitrary circuits.

Definition 5.5 (context width)**.** Work in the NNF of C (size $\le 2|C|$, nested gates flattened). For an $\wedge$-gate g let $S_g = \bigcup_{a \ne b \text{ children of } g} \mathrm{var}(a) \cap \mathrm{var}(b)$ be its overlap. For a gate h let $\mathrm{Up}(h)$ be the set of $\wedge$-gates from which h is reachable (including h), and define the *context* $\kappa(h) = \mathrm{var}(h) \cap \bigcup_{g \in \mathrm{Up}(h)} S_g$. The context width is $\mathrm{cw}(C) = \max_h |\kappa(h)|$.

Theorem 5.6 (context-width compilation)**.** *Every circuit C is equivalent to a DNNF of size $O(|C| \,\mathrm{cw}\, 2^{\mathrm{cw}})$, computable in the same time. $\mathrm{cw}(C) = 0$ iff the NNF of C is already a DNNF. If moreover every $\vee$-gate has semantically disjoint children, the result is a d-DNNF. Consequently (via Theorem 6.4(a) and [MdC26]) p has an FPRAS under every product distribution with rational marginals, running in time* $\mathrm{poly}(|C| 2^{\mathrm{cw}}, n, \max_i \log b_i, 1/\varepsilon, \log(1/\delta))$*; polynomial when* $\mathrm{cw} = O(\log |C|)$.

Proof. Two facts. (F1) If h is a child of g then $\mathrm{Up}(h) \supseteq \mathrm{Up}(g)$, so $\kappa(g) \cap \mathrm{var}(h) \subseteq \kappa(h)$. (F2) If h_i, h_j are distinct children of an $\wedge$-gate g then $\mathrm{var}(h_i) \cap \mathrm{var}(h_j) \subseteq S_g \cap \mathrm{var}(g) \subseteq \kappa(g)$. Nodes of the DNNF: for each gate h and $\sigma \in \{0,1\}^{\kappa(h)}$ a node $N[h, \sigma]$ for $h|_\sigma$; for each wire $g \to h$ and each assignment π to $\kappa(g) \cap \mathrm{var}(h)$ a node $M[h, \pi] = \bigvee_\tau (\mathrm{lits}(\tau) \wedge N[h, \pi \cup \tau])$, τ ranging over assignments to $\kappa(h) \setminus \mathrm{dom}\, \pi$ (a deterministic Shannon expansion; $\pi \cup \tau$ covers $\kappa(h)$ by F1). Set $N[g, \sigma] = \mathrm{op}_j M[h_j, \sigma|_{\kappa(g) \cap \mathrm{var}(h_j)}]$ for a gate g with children h_j, literal leaves become constants when their variable is in $\kappa(h)$, and the root is $M[\mathrm{root}, \emptyset]$. Correctness is induction on $g|_\sigma = \mathrm{op}_j h_j|_\sigma$. Decomposability: inside M, the literals of τ live on $\kappa(h) \setminus \mathrm{dom}\, \pi$ and $N[h, \cdot]$ on $\mathrm{var}(h) \setminus \kappa(h)$; at an $\wedge$-gate the children's variables lie in $\mathrm{var}(h_j) \setminus \kappa(g)$, pairwise disjoint by F2. Each wire contributes $O(\mathrm{cw}\, 2^{\mathrm{cw}})$ nodes. The d-DNNF claim holds because conditioning preserves disjointness and the M-nodes are deterministic. □

Example 5.7 ($\vee$'s are easy, $\wedge$'s are hard)**.** Let $C = \bigvee_{i \le t} F_i$ where each F_i is a CNF with clauses on sliding windows of width w along its own variable order. Then $\mathrm{cw} = O(w)$, so an FPRAS exists, while p can be $2^{-\Theta(n)}$. Exact computation is #P-hard already for ORs of $t = O(1)$ *read-once* CNFs: edge-colour a bounded-degree graph with $\Delta + 1$ colours, write its vertex-cover CNF as $\bigwedge_c \varphi_c$ with each φ_c a matching CNF, and recover $\#\bigwedge_c \varphi_c$ from $\#\bigvee_{c \in S} \varphi_c$, $S \subseteq [\Delta + 1]$, by inclusion–exclusion; counting vertex covers in graphs of degree ≥ 3 is #P-complete.

The same construction works with a *frontier* of gates that are compiled otherwise (small support by tabulation, bounded treewidth by [Dar09]): context width is then computed with frontier gates as leaves.

5.4 Modules and quotients

A gate g is a *module* if every path from an input in $\mathrm{var}(g)$ to the output passes through g. Then $C(x) = Q_g(g(x_{\mathrm{var}(g)}), x_{\mathrm{rest}})$ for the quotient Q_g obtained by replacing g with a fresh input, and under a product distribution $g(x)$ is independent of x_{rest}. Modules form a laminar family and can be found in $O(|C|\, n)$ time with dominators, as in fault-tree analysis [DR96].

Lemma 5.8 (sensitivity of quotients)**.** *Let $F(q) = \Pr_{y \sim \bigotimes_j \mathrm{Bern}(q_j)}[Q(y) = 1]$. If $\hat{q}_j \in e^{\pm \xi_j} q_j$ and $1 - \hat{q}_j \in e^{\pm \xi_j}(1 - q_j)$ for all j, then $F(\hat{q}) \in e^{\pm \sum_j \xi_j} F(q)$. If Q is monotone (resp. antitone) in y_j, the condition on q_j (resp. $1 - q_j$) alone suffices for that coordinate.*

Proof. F is multilinear: $F = q_j A + (1 - q_j) B$ with $A, B \ge 0$, so scaling either term by a factor in $e^{\pm \xi_j}$ scales F within $e^{\pm \xi_j}$. If Q is monotone in y_j then $A \ge B$, and for $c \ge 1$, $F(cq_j) = F + (c-1) q_j (A - B) \le F + (c-1) q_j A \le cF$; symmetrically for $c \le 1$. Iterate over coordinates. □

Walking the module tree bottom-up and estimating each module's probability with any sound method (exact for small quotients, the threshold dynamic program, DFSB, Monte Carlo for dense quotients) gives a sound estimator whose relative error is the sum of the modules' errors. Thresholds, majorities and other tractable functions of independent sub-circuits are the typical case: a threshold of 80 random modules with $p \approx 10^{-15}$ is estimated to 0.1% in seconds.

5.5 The presumption of independence, certified

Let $C = \bigwedge_{i=1}^{m} A_i$ be the top $\wedge$-tree of C with arbitrary sub-circuits A_i of supports S_i, $B_i = \neg A_i$, $b_i = D(B_i)$, and G the dependency graph ($i \sim j$ iff $S_i \cap S_j \neq \emptyset$). The presumption of independence estimates p by $M = \prod_i (1 - b_i)$. Call C *unate* if there is a sign pattern such that every path from x_v to the output has the same parity of negations; after flipping the negative variables, every B_i is increasing. This is checked syntactically in $O(|C| n)$ time.

Theorem 5.9 (certified independence, first order)**.** *Let C be unate. For any ordering of the events and $d_i = \sum_{j < i,\, j \sim i} D(B_i \wedge B_j)$,*

$$M \;\le\; p \;\le\; \prod_i (1 - b_i + d_i) \;\le\; M \exp\Big(\frac{\Delta}{1 - \max_i b_i}\Big), \qquad \Delta = \sum_{\{i,j\}\colon i \sim j} D(B_i \wedge B_j).$$

The following procedure is sound on every circuit: check unateness; estimate every b_i with $O((\mu + 1)\varepsilon^{-2})$ independent samples (median of $O(\log(1/\delta))$ repetitions for $\hat{M}$), estimate Δ by sampling dependent pairs, and output $[\hat{M} e^{-\varepsilon/4}, \hat{M} e^{3\varepsilon/4}]$ only if all $\hat{b}_i \le 1/3$ and $\hat{\Delta} \le \varepsilon/16$. It is complete when $\max_i b_i \le 1/4$ and $\Delta \le \varepsilon/64$, at a cost of $O\big(\sum_i |A_i| \cdot ((\mu + 1)\varepsilon^{-2} + |E(G)|\varepsilon^{-1}) \log(m/\delta)\big)$ evaluations, independent of p (which can be $e^{-\Theta(\mu)}$).

Proof. Write $p = \prod_i (1 - c_i)$ with $c_i = \Pr[B_i \mid F_{<i}]$, $F_{<i} = \bigwedge_{j<i} \neg B_j$ (nonzero by Harris unless some $b_j = 1$). B_i is increasing and $F_{<i}$ decreasing, so $c_i \le b_i$ (Harris), giving $p \ge M$. Split the earlier events into neighbours N and non-neighbours R of i (Boppana–Spencer [BS89]): B_i is independent of F_R (disjoint supports), so $c_i \ge \Pr[B_i F_N \mid F_R] \ge b_i - \sum_{j \in N} \Pr[B_i B_j \mid F_R] \ge b_i - d_i$, the last step by Harris ($B_i B_j$ increasing, F_R decreasing). Then $1 - b_i + d_i \le (1 - b_i) e^{d_i/(1-b_i)}$. For the procedure: the log of $\hat{M}$ deviates from $\log M$ by a sum of independent terms of total variance $O(\mu/N)$ plus a quadratic remainder of the same order, so $N = O((\mu + 1)/\varepsilon^2)$ and a median give $|\log \hat{M} - \log M| \le \varepsilon/4$; the pre-tests reject with high probability when $\max b_i > 1/2$ or $\Delta > \varepsilon/4$, and otherwise $p \in [M, M e^{\varepsilon/2}]$. Only M needs relative accuracy; Δ and $\max_i b_i$ need additive accuracy. □

For a unate CNF all quantities are products of marginals and the sandwich is deterministic (Theorem 4.7); for triangle-freeness of $G(N, q)$ at $N = 10^4$, $\mu = 10$, the window is $e^{\pm 0.023}$. The bound needs only *syntactic* structure (unateness, supports); semantic monotonicity or independence is coNP-hard to check and invisible to sampling.

Second order. Correcting by covariances, $M_2 = \prod_i (1 - b_i + C_i)$ with $C_i = \sum_{j<i, j \sim i} \operatorname{Cov}(B_i, B_j)$, the same Harris/Boppana–Spencer argument together with Bonferroni inequalities gives $|\log p - \log M_2| \le E_3$, where E_3 is a sum of third-order quantities (joint probabilities of connected triples, and singles times connected pairs). On triangle-freeness of $G(10^4, q)$ with $\mu = 100$ ($p \approx e^{-99}$) the first-order window is $e^{2.1}$ while the second-order one is $e^{\pm 0.055}$, in a regime where the local-lemma counting conditions fail. For principal up-sets Mousset, Noever, Panagiotou and Samotij [MNPS20] give a full hierarchy of such bounds in terms of joint cumulants; the statement for arbitrary monotone sub-circuits and its proof are in Appendix A.

Proposition 5.10 (pairwise data is not enough)**.** *Let all variables be Bernoulli(q), $q \le 1/2$. The "stars" $B_i = y \wedge z_i$ and the "triangle" $B_1 = z_1 z_2, B_2 = z_2 z_3, B_3 = z_3 z_1$ have the same dependency graph, the same $b_i = q^2$ and the same $D(B_i \wedge B_j) = q^3$; taking M' disjoint copies, $\log(p_{\text{stars}}/p_{\text{triangles}}) = M' \log\left(1 + \frac{q^3(1-q)}{1-3q^2+2q^3}\right) \ge 0.29\,\Delta$. So no certificate computed from single and pairwise probabilities can have a window narrower than $e^{0.29\Delta}$: Janson's $e^{\Delta/(1-b)}$ is optimal up to the constant in the exponent.*

Proof. $p_{\text{stars}} = (1-q) + q(1-q)^3$ and $p_{\text{triangles}} = (1-q)^3 + 3q(1-q)^2 = 1 - 3q^2 + 2q^3$; their difference is $q^3(1-q)$. The bound 0.29 is the minimum over $q \in (0, 1/2]$ of $\log(1 + q^3(1-q)/(1-3q^2+2q^3))/(3q^3)$, attained near $q = 0.28$. □

Beyond pairs. Two further certified routes need no monotonicity. (i) *Interface dynamic programming.* If Z contains every variable shared by two of the A_i, then $p = \sum_{z \in \{0,1\}^Z} D_Z(z) \prod_i a_i(z_{I_i})$ with $a_i(y) = \Pr[A_i \mid x_{I_i} = y]$, $I_i = S_i \cap Z$, computed by variable elimination along a tree decomposition of the interface hypergraph with Monte Carlo (or recursively estimated) factor tables; with interval arithmetic this is sound for every circuit, and it handles strong, extensive correlations (on a chain of 300 modules with $p \approx 3 \cdot 10^{-115}$, where independence is off by e^{64}, the log error was below 0.06). This is DFSB's cutset rule organised by a tree decomposition. (ii) *The Möbius expansion* $\log p = \sum_{T \text{ connected}} \mu(T)$, whose first-order terms are the presumption of independence and whose higher terms are log-interaction corrections computed from non-rare joint probabilities; a Kotecký–Preiss argument [KP86] certifies the tail when $\max_i b_i \le (2e^3 d)^{-r}$ (d the degree of G, r the number of events per variable), giving an FPRAS for bounded d.

5.6 Certified telescoping with Las Vegas samplers

Annealing estimators write $p = \prod_k r_k$ with $r_k = \Pr_{\pi_{k-1}}[\neg B_k]$ and $\pi_{k-1} = D \mid F_{<k}$, and estimate each ratio by sampling from π_{k-1}. They are sound only if the samples really come from π_{k-1}; Markov chains give no certificate of that. *Exact* Las Vegas samplers do, if their running time is certified empirically:

Proposition 5.11 (certified truncation)**.** *Let $\mathcal{A}$ halt with probability 1 and output exactly π, with running time τ. Truncated at T and conditioned on halting, its output law π_T satisfies $d_{\mathrm{TV}}(\pi_T, \pi) \le \Pr[\tau > T]$. If at most $\eta R/2$ of $R = \lceil 8 \ln(1/\delta)/\eta \rceil$ truncated runs fail, then $\Pr[\tau > T] \le \eta$ except with probability δ.*

Proof. $\pi = (1-\theta)\pi_T + \theta\pi_{>T}$ with $\theta = \Pr[\tau > T]$, so $\pi_T - \pi = \theta(\pi_T - \pi_{>T})$. If $\theta > \eta$, then by Chernoff at most $\eta R/2$ failures occur with probability at most $e^{-\eta R/8} \le \delta$. □

For unate C (flipped so that the B_i are increasing) the target $D \mid F$ is a distribution on a down-set, and heat-bath Glauber dynamics ($x_v \leftarrow 1$ iff $u < q_v$ and no event containing v fires on $x^{v \to 1}$) is anti-monotone. The bounding-chain coupling from the past of Häggström and Nelander [HN98, PW96] (lower chain updated with the upper chain's local check and vice versa) is then an exact Las Vegas sampler for $D \mid F_{<k}$, whatever the events are.

Theorem 5.12 (certified telescoping)**.** *For unate C, group the events into stages with Harris-guaranteed ratios $r_k \ge 1/2$, certify each stage's truncation at level $\eta = \varepsilon/(16m)$ by Proposition 5.11, and estimate each r_k with $O((\mu+1)/\varepsilon^2)$ truncated exact samples (median trick). The output is within $(1 \pm \varepsilon)p$ with probability $1 - \delta$ for every unate circuit and every truncation level; it is a verdict (not "fail") whenever coalescence is fast, which is provable when $\gamma = 1 - \max_v q_v\, r(k_{\max} - 1) > 0$ (r events per variable, supports of size $\le k_{\max}$), with $T = O((n/\gamma)\log(nm/\varepsilon))$.*

Proof sketch. Bias: each $r_k \ge 1/2$ is shifted by at most η in total variation, a relative shift of 2η; over m stages at most $\varepsilon/4$. Variance: as for products of independent estimates. Coalescence: the disagreement set shrinks in expectation by a factor $1 - \gamma/n$ per step. □

This covers, for instance, hard-core-type instances where Janson's Δ is extensive and the local-lemma conditions fail; beyond the provable regime the method stays sound and decides empirically whether it can answer. By Karp–Luby–Madras, ORs of unate $\wedge$-circuits (with different polarities) follow.

5.7 Specialised modules

Some structures are cheaper to detect and use directly. Each module below is sound on every input and recognises its class syntactically:

- *Bounded treewidth.* Variable elimination on the Tseitin factor graph along a greedy elimination order computes p exactly in time $O(|C|\, 2^{w+1})$ for elimination width w; polynomial for $w = O(\log |C|)$, and an order of width $O(\mathrm{tw})$ can be found in time $2^{O(\mathrm{tw})}|C|$ [Dar09, SS10, BDD+16].
- *Thresholds of independent blocks.* Exponential tilting (Proposition 4.6): $K \le \frac{4}{3}(2\sqrt{k}+1)$ for k unit-weight blocks; real weights under product distributions via the branching-program FPTAS of [GKM10] (see also [GKM+11]).

- *Unate circuits with sparse overlaps or fast coalescence.* Theorems 4.7, 5.9 and 5.12.
- *Small context width.* Theorem 5.6, with the #DNNF FPRAS or, in practice, the envelope estimator on the compiled DNNF.
- *Local-lemma regime.* k-CNFs (more generally, local CSPs) whose variables lie in at most $d \leq 2^{ck}$ constraints admit FPRASs [Moi19, FGYZ21, FHY21, JPV21, HWY22]; the best published constant is $d \lesssim 2^{k/4.82}$ [WY24]. The condition is syntactic.
- *Algebraic.* Affine and degree-2 constraints over $\mathbb{F}_2$ under the uniform distribution: exact counting and sampling by Gaussian elimination and Dickson's normal form [EK90]; usable as envelope pieces.

Theorem 5.13 (meta-procedure)**.** *Run DFSB and every applicable module, each at confidence δ/M, and output the intersection of their intervals (or "no verdict" if it is not tight). The result is sound on every circuit, runs in polynomial time, and is complete on the union of the classes above.*

Proof. With probability $1 - \delta$ all M intervals are valid simultaneously, and then so is their intersection; tightness of any one implies tightness of the intersection. □

5.8 What remains out of reach

Proposition 3.1 rules out beating sampling on all circuits; under the exponential-time hypothesis [IP01], no sound procedure that is complete on all circuits with n inputs runs in time $2^{o(n)}\mathrm{poly}(|C|)$, since on a 3-CNF ($p = 0$ or $p \geq 2^{-n}$) it would decide satisfiability; so no general procedure beats sampling by more than a factor $2^{o(\log(1/p))}$ on the satisfiable ones. Two natural kinds of rarity also defeat the syntactic tools:

- *Distributed surprise.* In a random 3-CNF with $m = \alpha n$ clauses, $1/3 < \alpha < \alpha_{\mathrm{sat}}$, every clause is satisfied with probability $7/8$ and the rarity comes from m overlapping weak constraints. Any drop-conjuncts relaxation keeps at most $n/3$ pairwise-disjoint clauses, so $D(E) \geq (7/8)^{n/3}$ while $p \leq (7/8)^m$: $K \geq (8/7)^{m-n/3}$, exponential. Annealing or multilevel-splitting estimators work well in practice here, but their accuracy rests on mixing of Markov chains that no polynomial-time test certifies, except where exact Las Vegas samplers exist (Theorem 5.12, for unate circuits): random 3-CNFs are not unate.
- *Pseudorandom rarity.* Preimages $[G(x) = y]$ of hash functions or PRGs: by Proposition 3.1(b), nothing beats sampling by more than a subexponential factor unless the primitive is broken.

6 With a prover: proof-carrying importance sampling

By Theorem 3.2 a verifier can certify an upper bound below the sampling scale only by reading the circuit, and by Theorem 3.4 no argument system does so for every circuit. The system below makes the deductive part as strong as any propositional proof system and leaves everything else to sampling. It is sound on every circuit and every argument, and complete exactly when the rarity of $C(x) = 1$ has a short *explanation*: a proof that the event is confined to a tractable region in which it is not rare.

6.1 The argument language

Fix a Cook–Reckhow proof system P for propositional unsatisfiability (resolution with RUP/DRAT/LRAT certificates, cutting planes, Frege, extended Frege, ...). For circuits F, G write $F \vdash_P G$ for a P-refutation of $\mathrm{Tseitin}(F) \wedge [F = 1] \wedge \mathrm{Tseitin}(G) \wedge [G = 0]$, which certifies $F \Rightarrow G$.

Definition 6.1 (PCIS arguments)**.** An argument is a finite DAG fixed before the verifier flips any coin. Every node v carries a circuit C_v (the root carries C) whose target is $q_v = D(C_v)$, and one rule:

Deductive rules (checked by the verifier, no randomness).
REFUTE: $C_v \vdash_P \bot$, so $q_v = 0$. EXACT: C_v is (or P-provably equivalent to) a d-DNNF, OBDD or SDD R, so $q_v = D(R)$ exactly. SUM: P-proofs of $C_v \Leftrightarrow \bigvee_i C_i$ and of $C_i \wedge C_j \vdash_P \bot$; then $q_v = \sum_i q_i$. PROD: $C_v = C_1 \wedge C_2$ with syntactically disjoint supports; then $q_v = q_1 q_2$. SPLIT: $q_v = (1 - q_x) q_{v|x=0} + q_x q_{v|x=1}$, and $C_v \Rightarrow \ell$ gives $q_v = D(\ell) q_{v|\ell}$. MONO: $C_v \vdash_P C_w$ gives $q_v \leq q_w$.

Statistical rules (each with a confidence budget δ_v and a sample size written in the argument).
ENVELOPE: an envelope Φ (Definition 4.1) with $C_v \vdash_P \mathrm{supp}\Phi$, estimated by Theorems 4.3 and 4.5 or by the FPRAS-guided sampler of Theorem 6.4. PROPOSAL: a lower bound from a proposal (Section 6.3). MC: plain Monte Carlo on C_v.

The verifier checks every certificate and that $\sum_v \delta_v \leq \delta$, evaluates the statistical nodes with fresh coins, combines intervals by interval arithmetic through the deductive rules (all of which are monotone in their arguments), and outputs the root interval. If a domination proof is missing or wrong, the node is downgraded to a lower bound (upper end $+\infty$).

Theorem 6.2 (soundness)**.** *For every circuit C, product distribution D and string π,* $\Pr[p \notin V(C, D, \pi)] \leq \delta$.

Proof. Each deductive rule states a true identity or inequality between the true targets, by soundness of P and independence of variable-disjoint sub-circuits under D; interval arithmetic with nonnegative endpoints is inclusion-monotone. Let B_v be the event that statistical node v returns an interval missing q_v. Conditionally on everything computed before v starts, v's target and sampling law are fixed and it uses fresh coins, so $\Pr[B_v \mid \text{history}] \leq \delta_v$ by Theorem 4.3 (resp. Theorems 4.5 and 6.4 and Proposition 6.6); a downgraded node fails only if its lower end exceeds q_v, which has the same bound. Hence $\Pr[\bigcup_v B_v] \leq \sum_v \delta_v \leq \delta$, and on the complement every interval, including the root's, contains its target. □

Adaptive choices inside a node (sequential stopping) are allowed; what must be fixed in advance is the set of nodes and their budgets, or a summable schedule $\delta_v = \delta\, 2^{-v}$.

Proposition 6.3 (relative-error calculus)**.** *If every statistical leaf returns an interval of ratio at most $1 + \eta$ and every derivation of the root from the leaves (one child at SUM, SPLIT; all children at PROD) uses at most d leaves, the root interval has ratio at most $(1 + \eta)^d$. With $\eta = \varepsilon/(2d)$ the ratio is at most $1 + \varepsilon$.*

Proof. Sums and convex combinations with exact weights do not increase the worst ratio of their summands; products multiply ratios. □

6.2 The two-sided workhorse: certified envelopes

When the envelope is an arbitrary DNNF E given by the prover, the derivation-weighted estimator of Theorem 4.3 can be needlessly slow: its relative variance is $\frac{D(E)}{p}\,\mathbb{E}[\varphi \mid E]\,\mathbb{E}[1/\varphi \mid C] - 1$, which depends on the *dispersion* of the number of derivations, and for $E = \bigwedge_{i \le m}(x_i \vee y_i)$ is $(10/9)^m - 1$ under every reweighting of the $\vee$-edges, although $D(E) = (3/4)^m$ is known exactly. The Meel–de Colnet FPRAS for DNNF [MdC26] removes the dispersion, through a Knuth-style sequential sampler whose weights are exactly unbiased whatever errors the counter makes.

Theorem 6.4 (FPRAS-guided envelope sampling)**.** *Let E be a DNNF and D a product distribution with rational marginals.*

(a) (Weighted counting.) Smoothing E and replacing each literal by a comparator on $\lceil \log_2 b_i \rceil$ fresh bits gives a DNNF E' with $\#E' = (\prod_i b_i)D(E)$; so the FPRAS approximates $D(E \mid \rho)$ for every partial assignment ρ.

(b) (Sampler.) Draw $x_1, \dots, x_n$ in turn: with counter outputs $A_c \approx D(E \mid \rho, x_i = c)$ (zero iff $E \mid \rho, x_i = c$ is unsatisfiable, which is decidable in linear time), set $x_i = c$ with probability $D_i(c)A_c/S$, $S = \sum_c D_i(c)A_c$, and multiply the weight w by S/A_{x_i}. Then for every circuit C and every *law of the counter's errors, $\mathbb{E}[C(x)\,w] = D(C \wedge E)$.*

(c) (Cost.) With counter accuracy $1/(4n)$ and confidence $\varepsilon/(16n)$ per call, $w \in [0.58, 1.70]\cdot D(E)$ outside an event of weighted mass $(\varepsilon/8)D(C \wedge E)$; clipping at $3\hat{D}(E)$ and running the stopping rule gives a $(1 \pm \varepsilon)$ estimate of $D(C \wedge E)$ with $O(K \log(1/\delta)/\varepsilon^2)$ samples, $K = D(E)/D(C \wedge E)$, each costing one evaluation of C and $2n$ counter calls.

Proof sketch. (a) Smoothing makes every derivation mention every variable exactly once, so the substituted comparator blocks are all constrained; then each x has $(\prod_i b_i)D(x)$ preimages. (b) At each step the probability of choosing c times the weight factor is $D_i(c)$, whatever A_0, A_1 are (Knuth's estimator [Knu75]); pre-drawing the counter outputs for every prefix and summing over paths gives $\mathbb{E}[C(x)w] = \sum_x D(x)C(x)\mathbf{1}_E(x)$ (the walk only enters satisfiable prefixes). (c) On the event that all $2n$ counts on the path are accurate, the weight telescopes to $D(E)/D(E \mid x) = D(E)$ up to $e^{\pm 2.1/(4n)}$ per step; apply (b) to the indicator of the bad event to bound its contribution, then the stopping rule [DKLR00]. □

Theorem 6.5 (certified envelopes)**.** *Let $\mathcal{R}$ be the closure under decomposable products and finite unions of subcubes, OBDDs, SDDs, d-DNNFs and, under the uniform distribution, affine subspaces, together with DNNFs via Theorem 6.4. Let $\mathcal{K}_P(m, K)$ be the class of circuits C for which there are $E_1, \dots, E_r \in \mathcal{R}$ and a P-proof $C \vdash_P \bigvee_j E_j$, all of total size at most m, with $\sum_j D(E_j) \le K\,p$. The PCIS verifier is sound on all circuits and complete on $\mathcal{K}_P(m, K)$, in time $\mathrm{poly}(m, |C|) + O(K \log(1/\delta)/\varepsilon^2) \cdot (|C| + m)$ (plus FPRAS overhead for DNNF pieces). Without the proof the same samples still certify a lower bound.*

Proof. Soundness is Theorem 6.2. For completeness, sample $j \propto D(E_j)$, $x \sim D \mid E_j$, and set $W = C(x)\Lambda/\mathrm{cov}(x)$ with $\Lambda = \sum_j D(E_j)$ and $\mathrm{cov}(x) = |\{j : x \in E_j\}|$ (Karp–Luby–Madras [KLM89]). Then x has density $D(x)\mathrm{cov}(x)/\Lambda$, $0 \le W \le \Lambda$, and $\mathbb{E}W = D(C \wedge \bigcup_j E_j)$, which is p under the proved domination. Apply Theorem 4.3 with $Z = \Lambda$. □

Theorem 6.5 lifts every tractable counting class from *syntactic* envelopes (Section 5) to *provable* ones. Examples:

- *Hidden triggers.* $C = R \wedge T$ where T is a test that holds only on a subcube $x_A = a$ but is not written as one: through a linear hash $[Mx_A = Ma]$ over $\mathbb{F}_2$, through two differently built $\oplus$-trees for the same parities, through a multiplier. A SAT solver produces a refutation of $C \wedge (x_A \neq a)$; with $K = 1/\Pr[R \mid x_A = a]$ the verifier needs $O(K/\varepsilon^2)$ samples, Monte Carlo $2^{|A|}K/\varepsilon^2$. Resolution needs exponential size for dense linear hashes, so here the proof system matters (Section 7); extended resolution, or an affine envelope, handles them in polynomial size.
- *Planted unions.* $C = \bigvee_j (T_j \wedge G_j)$, T_j subcubes of large codimension, G_j dense: $K \leq (\text{overlap})/\min_j \Pr[G_j \mid T_j]$.
- *Thresholds with side conditions.* $C = [w \cdot x \geq \theta] \wedge G$ with integer w: a cutting-planes proof of $w \cdot x \geq \theta$ from C and the soft envelope $\beta^{w \cdot x - \theta}$ ($\beta > 1$ rational) give $K = O(\sqrt{n})$ (Proposition 4.6); numerically $K = 37$ at $n = 800$, $p = 4 \cdot 10^{-125}$.
- *Hidden independence.* A proof that $C \Leftrightarrow \bigwedge_j C_j(x_{S_j})$ with disjoint S_j turns a product of k moderately rare events into k separate estimations (PROD).

Sandwiches. A certificate may also give, for every gate g, OBDDs $L_g \subseteq g^{-1}(1) \subseteq U_g$ in a fixed variable order, checked gate by gate ($L_{a \wedge b} \Rightarrow L_a \wedge L_b$, $U_a \wedge U_b \Rightarrow U_{a \wedge b}$, $L_{\neg a} \wedge U_a \equiv \bot$, $\neg L_a \Rightarrow U_{\neg a}$, and dually for $\vee$), each check an OBDD operation. Then $p \in [D(L_{\text{out}}), D(U_{\text{out}})]$ deterministically, and sampling $D \mid (U_{\text{out}} \setminus L_{\text{out}})$ estimates the gap with $O(\frac{D(U \setminus L)}{p} \varepsilon^{-2} \log \frac{1}{\delta})$ samples. No SAT proof is needed.

6.3 Lower bounds need no deduction

Proposition 6.6 (trace-reconstructible proposals)**.** *The prover supplies a seed law P_R on $\{0,1\}^r$ with computable pmf and an exact sampler, a generator $\psi : \{0,1\}^r \to \{0,1\}^n$, an inverter $\phi : \{0,1\}^n \to \{0,1\}^r$, and a clip level τ. For $R \sim P_R$ and $X = \psi(R)$ let $W = \mathbf{1}[\phi(X) = R]\, C(X) \min(D(X)/P_R(R), \tau) \in [0, \tau]$. Then for every C and every choice of the other objects,*

$$\mathbb{E}W = \sum_{x \in C^{-1}(1)\colon \psi(\phi(x)) = x} \min\big(D(x), \tau P_R(\phi(x))\big) \leq p,$$

so the lower end of a KL interval for $\mathbb{E}W$ is a sound lower bound. It is within $1 - \varepsilon$ of p after $O(K \log(1/\delta)/\varepsilon^2)$ samples if $\tau \leq Kp$ and the pairs x with $D(x) \leq \tau P_R(\phi(x))$ and $\psi(\phi(x)) = x$ carry $(1 - \varepsilon/2)p$.

Proof. ψ is injective on $R^* = \{r : \phi(\psi(r)) = r\}$ (if $\psi(r) = \psi(r') = x$ then $r = \phi(x) = r'$) and maps it onto $\{x : \psi(\phi(x)) = x\}$; substitute $x = \psi(r)$. Without the trace check each x would be counted once per preimage and $\mathbb{E}W$ could exceed p. □

Special cases: autoregressive models or probabilistic circuits ($r = n$, $\psi = \phi = \text{id}$, $P_R = Q$), witness lists (exact lower bound $\sum_i D(x^{(i)})$, no sampling), an unproved envelope, and generators with inverters such as normalising flows.

6.4 Surprise accounting, made exact

Write $\sigma = \log_2(1/p)$ for the surprise of the event. An envelope argument explains $\sigma_e = \log_2(1/D(E))$ bits deductively and leaves $\sigma_u = \log_2(D(E)/p)$ bits to sampling:

$$\text{verifier time} \;=\; \text{poly}(|\pi|, |C|) \;+\; O\big(2^{\sigma_u} \varepsilon^{-2} \log(1/\delta)\big) \cdot (|C| + |E|).$$

For a skeleton of independent components (PROD at the top) the second term is $\sum_\lambda 2^{\sigma_u(\lambda)}$ rather than $2^{\sum_\lambda \sigma_u(\lambda)}$: unexplained surprise spread over t independent components costs about $t\,2^{\sigma_u/t}$. Both terms are necessary in the black-box sense:

Theorem 6.7 (proof bits and surprise bits)**.** *Let a verifier access C only through T evaluations (adaptive, from any distributions) and read an ℓ-bit argument; let D be uniform.*

(a) If it is δ-sound for lower bounds on all circuits and, for each subcube S of codimension k, some argument makes it output $L > 0$ on $\mathbf{1}_S$ with probability $1-\delta$, then $T \cdot 2^\ell \geq (1-2\delta)2^k$. Each bit of argument buys at most one bit of surprise, and this is tight: revealing j coordinates of S leaves $T = O(2^{k-j}\varepsilon^{-2}\log(1/\delta))$.

(b) Inside a certified envelope E, distinguishing $C_0 = E \wedge [x_I = b]$ from $C_1 = E \wedge ([x_I = b] \vee [x_{I'} = b'])$ (factor 1.5 apart) needs $T \geq (1-2\delta)D(E)/p$ evaluations: the 2^{σ_u} term cannot be avoided by any method that treats the unexplained part of C as a black box.

Proof. (a) Couple the runs on $\mathbf{1}_S$ and on the zero function with the same coins; they differ only if a query lands in S, which for a fixed query has probability 2^{-k} over a uniformly random S of the family. Soundness on the zero function forces the event "some query lands in S" to have probability at least $1-2\delta$ under the argument π_S; averaging over S and summing over the at most 2^ℓ arguments gives $1-2\delta \leq 2^\ell T 2^{-k}$. (b) The same coupling, with C_0 in place of the zero function and the random subcube $[x_{I'} = b']$ planted. □

So the cost of certifying a rare event is governed by $\min_\pi \left(|\pi| + 2^{\sigma_u(\pi)}\right)$, a two-part, MDL-like cost of the kind that "surprise accounting" [Hil24] proposes for heuristic explanations, here with a sound verifier and a matching lower bound.

6.5 How much stronger is the prover?

A universal verifier. Fix a consistent theory T (say ZFC). Let a V_T-argument be a randomized program A, a time bound t, a budget $\delta_A \leq \delta$, and a T-proof of the bounded arithmetic sentence "at most $\delta_A 2^t$ coin strings make $A^{(t)}(C, D)$ output an interval missing $\sum_x D(x)C(x)$"; V_T checks the proof and runs A. It is sound whenever T is (a false bounded sentence has a true Σ_1 negation, which T would refute), and it p-simulates every argument system whose soundness T proves, by instantiating that proof. So it can use any FPRAS, compiler, mixing-time bound or combinatorial theorem as a certified rule. This is the analogue, for estimation, of proof systems that are optimal relative to a theory [KP89].

Lower bounds: the prover helps unless NP $\subseteq$ BPP. Uniquely satisfiable circuits have $p = 2^{-n}$, certified exactly by one witness. A sound Task-1 procedure that outputs any $L > 0$ on all of them decides SAT via Valiant–Vazirani [VV86]. Under one-way permutations (with a Goldreich–Levin hard-core bit [GL89]), $C_y(x, r) = [(\pi(x), r, \langle x, r\rangle) = y]$ gives an average-case version with $(1+\varepsilon)$-tight certificates.

Upper bounds: the prover helps exactly when P is not weakly automatable. For a refutation system P let Z_P^m be the circuits C with a P-refutation of size $m(|C|)$. A sound Task-1 procedure that outputs $[0, 0]$ on all of Z_P^m exists iff the canonical pair (Z_P^m, SAT) is separable in randomized polynomial time (if it outputs $[0, 0]$, the circuit cannot be satisfiable, by

soundness; conversely run the separator); separability of the canonical pair is weak automatability of P [Pud03]. Padding turns this into a separation for positive, exactly certified p: with $\mathrm{pad}_a(C)(\beta, x, w) = (\beta \wedge x = 0^n \wedge w = 0^a) \vee (\neg\beta \wedge C(x))$, $p = 2^{-n-a-1}$ exactly when C is unsatisfiable (certified by one point and P's refutation), while $p \geq 2^{-n-1}$ when C is satisfiable, so any Task-1 procedure complete within a factor 2^a separates the canonical pair. Known results then give:

Corollary 6.8. *If factoring Blum integers is hard for polynomial-size circuits, then for $P \in \{\mathrm{TC}^0\text{-Frege}, \text{Frege}\}$ [BPR00], and for extended Frege if RSA is secure [KP98], there are circuit families on which the PCIS verifier with P certifies p exactly while no sound polynomial-time procedure without a prover is complete, even within a factor 2^a. Under LWE the same holds against quantum procedures for TC^0-Frege [ACG24].*

Finding the short proofs is the hard part: even for resolution, finding a refutation of size polynomial in the shortest one is NP-hard [AM20]. And for preimage circuits $[G(x) = y]$ of pseudorandom generators, short proofs that $y \notin \mathrm{range}(G)$ are conjectured not to exist in strong proof systems (proof complexity generators [ABSRW04, Kra04]), so there the verifier cannot beat sampling for upper bounds either.

Two-sided, exact. Let $L \in \mathsf{UP} \cap \mathsf{coNP}$ with verifiers A (unique witness, 0^k never a witness) and B, and $C_u(\beta, v) = (\beta \wedge v = 0^k) \vee (\neg\beta \wedge A(u, v))$, so $p(C_u) = 2^{-k-1}(1 + [u \in L])$. With T proving uniqueness and disjointness, V_T certifies both values exactly (by the witness of A or of B), while a Task-1 procedure complete on $\{C_u\}$ decides L. For bits of the prime factorisation (AKS-certified primes) this gives a two-sided separation under "factoring $\notin$ BPP".

Why not NP-hardness? If SAT reduces (randomised, many-one) to the gap problem "$p \geq \gamma t$ versus $p \leq t$" with unsatisfiable formulas mapped to the *small* side, and some MA system is complete for upper bounds on that side, then $\mathsf{coNP} \subseteq \mathsf{AM}$: Arthur samples the reduction, Merlin sends arguments, Arthur accepts if most of them certify $U < \gamma t$. With the opposite orientation, Goldwasser–Sipser already puts the problem in AM. Either way PH collapses [BHZ87]. So upper-bound and two-sided separations must come from structured, non-NP-hard assumptions (factoring, RSA, LWE), as above, while lower-bound separations may come from $\mathsf{NP} \not\subseteq \mathsf{BPP}$.

Universality. Upper bounds: complete for all circuits only if $\mathsf{coNP} \subseteq \mathsf{MA}$ (Theorem 3.4). Lower bounds: complete for all circuits within a constant factor iff prSBP $\subseteq$ prMA [BGM06]; this fails relative to an oracle, and holds under derandomisation hypotheses strong enough to fool nondeterministic circuits, with a verifier whose soundness then depends on the hypothesis.

7 Experiments

The package `rareprob` (Python, numpy) implements everything above: bit-parallel circuit evaluation, NNF conversion with small-cone canonicalisation, the envelope compiler and estimator with boxes, DFSB, variable elimination, exponential tilting, the Harris–Janson sandwich, Gaussian-elimination forcing, a two-watched-literal DRUP checker, and the PCIS verifier for envelope arguments (with an honest prover that calls a CDCL solver with proof logging). All intervals are Hoeffding–KL intervals combined by union bounds, exactly as in the proofs. Monte Carlo cost is not simulated for tiny p: we report the number of samples its KL interval needs, at the same

(ε, δ), when the empirical mean equals p (this favours Monte Carlo). Scripts and raw results are in `experiments/`; every number below is reproduced by them.

Soundness (Table 1). We ran each procedure many times at $\delta = 0.1$ on random circuits and on families with exactly known p (by enumeration, inclusion–exclusion or dynamic programming), and counted intervals that miss p. Adversarial arguments were included: a wrong envelope (a dropped term) with the honest proof of the right envelope, a wrong envelope with a list of random clauses as "proof", and an envelope disjoint from most of C's mass with no proof. Every miss rate is below δ, as it must be; most are 0, because the union bounds and KL intervals are conservative.

procedure	trials	misses	miss rate
envelope (auto) on random circuits	300	0	0.000
DFSB on random circuits	150	2	0.013
KLM envelope on DNFs	300	0	0.000
envelope with boxes on block products	100	0	0.000
exponential tilting on thresholds	200	0	0.000
Harris-Janson sandwich (deterministic)	300	0	0.000
argument: honest envelope + DRUP proof	150	0	0.000
argument: wrong envelope + stale proof	150	0	0.000
argument: wrong envelope + bogus proof	150	0	0.000
argument: disjoint envelope, no proof	150	0	0.000

Table 1: Empirical miss rates at $\delta = 0.1$ (`exp_coverage.py`).

Cost versus rarity (Figure 1, Table 2). Four families with exactly known p, at $\varepsilon = 0.1$, $\delta = 0.05$:

- *obfuscated DNF*: 16 random terms of width w over 64 variables, every literal routed through $(x \oplus r) \oplus r$ gadgets, terms built as trees, the $\vee$ written as $\neg \wedge \neg$. Small-cone canonicalisation recovers the literals and the envelope is the DNF itself ($K \approx 1$);
- *triggers*: $R(x_{\text{body}}) \wedge [x_A = a]$ with the test written as $\neg(x_i \oplus a_i)$ gates and R a dense random circuit; forcing finds $x_A = a$;
- *products of blocks*: $\bigwedge_{j \le t} B_j$ with B_j random 12-gate circuits on disjoint 6-variable blocks (dense leaves);
- *thresholds of blocks*: $[\sum_j B_j \ge t]$ over k random blocks, by exponential tilting.

The cost of Monte Carlo grows like 2^σ in the surprise $\sigma = \log_2(1/p)$; ours stays polynomial: flat for DNFs and triggers, cubic in the number of blocks for products (the d^2 accuracy split of Theorem 5.2), and $O(\sqrt{k})$ for thresholds.

Context width (Table 3). ORs of t CNFs with clauses on sliding windows of width w, each CNF along its own random variable order, built as left-deep $\wedge$-chains. The compiler of Theorem 5.6 finds context width w and produces an equivalent DNNF; the envelope estimator on it needs only the minimum batch of 1,024 samples, because the top $\vee$'s children are rare and almost disjoint ($K \approx 1$). Instances with $n \le 24$ agree with enumeration.

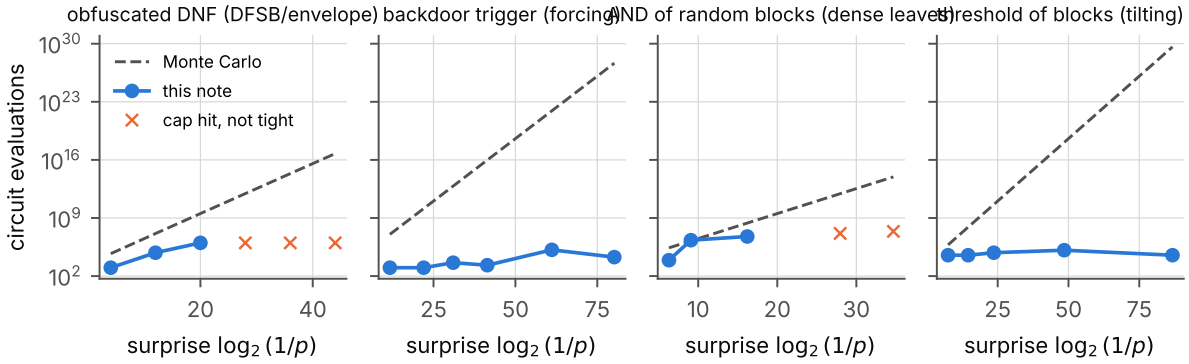


Figure 1: Circuit evaluations to reach an interval of ratio ≤ 1.1 ($\varepsilon = 0.1$, $\delta = 0.05$) against the surprise $\log_2(1/p)$, compared with Monte Carlo's requirement (`exp_speedup.py`).

Certified independence (Table 4). Unate random k-CNFs with up to $2 \cdot 10^5$ clauses: the Harris–Janson interval is deterministic, takes seconds to a few minutes in Python, and on 200 small instances it contained the exact p every time. At $n = 2{\cdot}10^6$, $p \in [6.2, 7.3]{\cdot}10^{-22}$; Monte Carlo would need about 10^{24} evaluations for the same ratio. The second-order sandwich of Appendix A was checked against enumeration on 382 random instances with no violations.

Certified telescoping (Table 5). Hard-core ladders: $x \sim \mathrm{Bern}(0.3)^n$ and $C =$ "x is an independent set of the $2 \times L$ ladder". Janson's Δ is extensive, so the sandwich is useless; bounding-chain CFTP gives exact samples, the truncation certificates contribute a bias below 1%, and the certified intervals contain the exact p (from variable elimination). In pure Python the constants are poor (each CFTP run costs thousands of updates), so for these moderate p Monte Carlo is still cheaper; the point is that the cost grows polynomially in L while $1/p$ grows exponentially, and that the method is sound without any mixing-time proof.

What the prover adds (Table 6, Figure 2). Hidden triggers $R \wedge T(x_S)$ whose test accepts only $x_S = a$: *linear-hash* tests $[Mx_S = Ma]$ with a dense $(k+4) \times k$ matrix over $\mathbb{F}_2$, and *SPN* tests $[\mathrm{SPN}(x_S) = \mathrm{SPN}(a)]$ with random bijective 4-bit S-boxes and dense invertible linear layers. Without a prover, Gaussian-elimination forcing recognises linear hashes, but nothing in Part 1 recognises SPN triggers, and DFSB returns an uninformative interval. With a prover, the argument is the subcube $x_S = a$ plus a DRUP proof (from Glucose) that $C \Rightarrow (x_S = a)$; the verifier checks it with its own RUP checker and samples only R. Resolution proofs grow exponentially with k even for linear hashes (resolution cannot do Gaussian elimination efficiently), which shows concretely that the class of Theorem 6.5 depends on the proof system.

figure pending

Figure 2: DRUP proof size for hidden triggers.

figure pending

Figure 3: Surprise accounting: samples needed when an argument reveals all but σ_u bits of a 40-bit trigger.

family	param	p	interval ratio	cost	Monte Carlo	speed-up
dnf-obfuscated	8	$6.08 \cdot 10^{-2}$	1.032	$1.02 \cdot 10^{3}$	$5.02 \cdot 10^{4}$	49.1
dnf-obfuscated	16	$2.44 \cdot 10^{-4}$	1.073	$6.55 \cdot 10^{4}$	$1.33 \cdot 10^{7}$	$2.03 \cdot 10^{2}$
dnf-obfuscated	24	$9.54 \cdot 10^{-7}$	1.083	$1.00 \cdot 10^{6}$	$3.41 \cdot 10^{9}$	$3.41 \cdot 10^{3}$
dnf-obfuscated	32	$3.73 \cdot 10^{-9}$	1.808	$1.00 \cdot 10^{6}$	$8.73 \cdot 10^{11}$	$8.73 \cdot 10^{5}$
dnf-obfuscated	40	$1.46 \cdot 10^{-11}$	270695.667	$1.00 \cdot 10^{6}$	$2.23 \cdot 10^{14}$	$2.23 \cdot 10^{8}$
dnf-obfuscated	48	$5.68 \cdot 10^{-14}$	inf	$1.00 \cdot 10^{6}$	$5.72 \cdot 10^{16}$	$5.72 \cdot 10^{10}$
trigger	10	$3.05 \cdot 10^{-4}$	1.087	$1.02 \cdot 10^{3}$	$1.06 \cdot 10^{7}$	$1.04 \cdot 10^{4}$
trigger	20	$2.38 \cdot 10^{-7}$	1.004	$1.02 \cdot 10^{3}$	$1.36 \cdot 10^{10}$	$1.33 \cdot 10^{7}$
trigger	30	$4.66 \cdot 10^{-10}$	1.091	$4.10 \cdot 10^{3}$	$6.98 \cdot 10^{12}$	$1.70 \cdot 10^{9}$
trigger	40	$3.41 \cdot 10^{-13}$	1.087	$2.05 \cdot 10^{3}$	$9.53 \cdot 10^{15}$	$4.65 \cdot 10^{12}$
trigger	60	$4.00 \cdot 10^{-19}$	1.066	$1.45 \cdot 10^{5}$	$8.13 \cdot 10^{21}$	$5.61 \cdot 10^{16}$
trigger	80	$7.24 \cdot 10^{-25}$	1.029	$1.92 \cdot 10^{4}$	$4.49 \cdot 10^{27}$	$2.34 \cdot 10^{23}$
blocks	4	$1.27 \cdot 10^{-2}$	1.082	$8.19 \cdot 10^{3}$	$2.53 \cdot 10^{5}$	30.9
blocks	8	$1.85 \cdot 10^{-3}$	1.029	$2.11 \cdot 10^{6}$	$1.75 \cdot 10^{6}$	0.829
blocks	16	$1.33 \cdot 10^{-5}$	1.066	$5.88 \cdot 10^{6}$	$2.45 \cdot 10^{8}$	41.7
blocks	24	$3.86 \cdot 10^{-9}$	1.115	$1.38 \cdot 10^{7}$	$8.43 \cdot 10^{11}$	$6.12 \cdot 10^{4}$
blocks	32	$3.62 \cdot 10^{-11}$	1.361	$2.44 \cdot 10^{7}$	$8.98 \cdot 10^{13}$	$3.69 \cdot 10^{6}$
threshold	50	$5.59 \cdot 10^{-3}$	1.083	$3.28 \cdot 10^{4}$	$5.78 \cdot 10^{5}$	17.6
threshold	100	$4.10 \cdot 10^{-5}$	1.087	$3.28 \cdot 10^{4}$	$7.92 \cdot 10^{7}$	$2.42 \cdot 10^{3}$
threshold	200	$7.74 \cdot 10^{-8}$	1.089	$6.55 \cdot 10^{4}$	$4.20 \cdot 10^{10}$	$6.41 \cdot 10^{5}$
threshold	400	$2.51 \cdot 10^{-15}$	1.079	$1.31 \cdot 10^{5}$	$1.29 \cdot 10^{18}$	$9.87 \cdot 10^{12}$
threshold	800	$7.79 \cdot 10^{-27}$	1.096	$3.28 \cdot 10^{4}$	$4.17 \cdot 10^{29}$	$1.27 \cdot 10^{25}$

Table 2: The runs of Figure 1. "Cost" counts all circuit evaluations, including pilots and box estimation.

n	t	w	cw	$\lvert C\rvert$	DNNF size	interval for p	samples	time (s)
20	4	3	3	249	640	[0.39, 0.416] (p =0.393)	1024	0.1
24	4	3	3	304	627	[0.209, 0.221] (p =0.215)	1024	0.1
100	4	3	3	1375	2350	$[9.50 \cdot 10^{-6}, 9.54 \cdot 10^{-6}]$	1024	0.6
200	4	3	3	2757	4575	$[1.57 \cdot 10^{-11}, 1.58 \cdot 10^{-11}]$	1024	0.8
400	4	3	3	5633	8927	$[2.91 \cdot 10^{-23}, 2.92 \cdot 10^{-23}]$	1024	2.1
800	8	3	3	22300	33927	$[3.93 \cdot 10^{-46}, 3.95 \cdot 10^{-46}]$	1024	10.8
400	4	5	5	7115	15137	$[1.46 \cdot 10^{-5}, 1.46 \cdot 10^{-5}]$	1024	6.0

Table 3: Context-width compilation of ORs of window-CNFs (`exp_ctx.py`).

Surprise accounting (Figure 3). For a 40-bit trigger, an argument that reveals $40 - \sigma_u$ of the bits (a subcube envelope on those bits) leaves σ_u bits to sampling; the number of samples grows as 2^{σ_u}, as Theorem 6.7 says it must.

8 Discussion and open problems

What the answers are. Without a prover, one can beat sampling exactly where the rarity of $C(x) = 1$ is *visible*: built from independent parts, forced literals, small cutsets, unions, thresholds of independent blocks, or sparse monotone overlaps, possibly disguised by local rewriting. DFSB plus the specialised modules is sound on every circuit and costs poly$\cdot K$ on the circuits whose ideal K-explanation tree is small (Theorems 5.2 and 5.13). With a prover, the visible structure may be replaced by *provable* structure: anything a propositional proof system can certify about where the rare inputs live (Theorem 6.5). In both settings the cost has the same shape, polynomial in the explained surprise and exponential in the unexplained surprise, and the exponential part is unavoidable for black-box residuals (Theorem 6.7).

n	clauses	width	certified interval	ratio	time (s)
10000	2000	10	[0.142, 0.147]	1.039	0.3
100000	20000	12	$[7.57 \cdot 10^{-3}, 7.84 \cdot 10^{-3}]$	1.035	6.3
200000	60000	12	$[4.34 \cdot 10^{-7}, 5.06 \cdot 10^{-7}]$	1.167	30.8
1000000	200000	14	$[4.99 \cdot 10^{-6}, 5.14 \cdot 10^{-6}]$	1.030	113.8
2000000	200000	12	$[6.19 \cdot 10^{-22}, 7.35 \cdot 10^{-22}]$	1.187	42.0

Table 4: Harris–Janson sandwich for random unate CNFs (`exp_janson.py`).

L	n	p (exact)	Harris–Janson window	certified telescoping	CFTP runs
20	40	$2.31 \cdot 10^{-2}$	$[4.21 \cdot 10^{-3}, 0.117]$	$[1.98 \cdot 10^{-2}, 2.79 \cdot 10^{-2}]$	$2.34 \cdot 10^{5}$
40	80	$4.90 \cdot 10^{-4}$	$[1.47 \cdot 10^{-5}, 1.43 \cdot 10^{-2}]$	$[4.11 \cdot 10^{-4}, 6.63 \cdot 10^{-4}]$	$4.42 \cdot 10^{5}$
60	120	$1.04 \cdot 10^{-5}$	$[5.12 \cdot 10^{-8}, 1.76 \cdot 10^{-3}]$	$[7.69 \cdot 10^{-6}, 1.41 \cdot 10^{-5}]$	$6.76 \cdot 10^{5}$

Table 5: Certified telescoping on hard-core ladders (`exp_cftp.py`).

Relation to heuristic arguments. ARC's programme [CNX22, Hil24] asks for estimators that use a model's internals to beat sampling, and measures explanations by surprise accounting. The present results isolate the part of that programme that can be made *sound against adversarial circuits*:

- the presumption of independence is certified where independence is syntactic (variable-disjoint parts, PROD) or where correlation inequalities control the error (Theorem 4.7);
- surprise accounting becomes an exact cost formula, $\mathrm{poly}(|\pi|) + 2^{\sigma_u(\pi)}$, with a matching black-box lower bound;
- what soundness forbids is equally sharp: upper bounds must come from deduction (Theorem 3.2), and no system certifies them for all circuits (Theorem 3.4). A heuristic estimator that is accurate on random circuits but not sound on adversarial ones is therefore not a counterexample to anything here; it lives in the gap between "no certificate exists" and "the estimate is wrong".

The rarity that defeats certification is *distributed surprise*: many weak, overlapping constraints (random CNFs below the threshold), or pseudorandomness. Annealing and splitting estimators handle the first in practice, and ARC-style heuristic estimators may handle both on typical instances; neither comes with a certificate that a polynomial-time verifier can check.

Open problems.

1. *Envelope lower bounds for distributed surprise.* Does every polynomial-size DNNF envelope of a random 3-CNF with $1/3 < \alpha < \alpha_{\mathrm{sat}}$ have $D(E)/p = 2^{\Omega(n)}$? Proved here only for drop-conjunct relaxations.
2. *Certified annealing.* Is there any polynomial-size, MA-checkable certificate that a sampler for $D \mid A_1 \wedge \cdots \wedge A_j$ is accurate (a checkable mixing certificate)? By Theorem 3.2 it cannot be purely statistical. This is what would make multilevel splitting sound.
3. *Nested unions over approximate leaves.* Can the DNNF FPRAS be run over DNNF skeletons whose leaves are sets given only by approximate counters, approximate samplers and membership tests? This would give unions of unions of products of dense blocks at any depth.

(table pending)

Table 6: Hidden triggers with and without a prover (`exp_arguments.py`). Part 1 ratio is that of the DFSB interval (∞: lower end 0); Part 2 ratio is that of the verifier's interval after checking the proof.

4. *A natural two-sided separation.* Find a $\mathsf{UP} \cap \mathsf{coUP}$ problem not in BPP (under a standard assumption) whose unique-witness property has polynomial-size extended Frege proofs; then the two-sided separation holds with a standard proof checker instead of a theory like ZFC.

5. *Optimal arguments.* How well can $\min_\pi \left(|\pi| + 2^{\sigma_u(\pi)}\right)$ be approximated for natural certificate languages? Choosing which conjuncts to keep is weighted set packing on supports; choosing branch variables is related to pseudo-tree depth.

6. *Networks.* Binarised or ReLU networks with rare activation patterns are the case of most interest for alignment. Interval-bound propagation already gives gate-wise outer approximations; can it be turned into sandwich certificates (Section 6) with small looseness for trained networks?

7. *Heuristic to certified.* When a presumption-of-independence estimate is accurate, when can it be compiled into a certificate with polynomial overhead, for instance by proving the independence it presumes, or by bounding the correlations it ignores?

A The second-order independence sandwich

Setting of Theorem 5.9: increasing events $B_1, \dots, B_m$ (after the global flip) with syntactic supports S_i, a product distribution, and a fixed order. For each i let $N_i = \{j < i : j \sim i\}$, $R_i = \{j < i : j \not\sim i\}$, $\Gamma(j)$ the neighbourhood of j, $b_i = D(B_i)$, $\beta_i = \sum_{j \in N_i} b_j$ (assumed < 1),

$$C_i = \sum_{j \in N_i} \mathrm{Cov}(B_i, B_j) \quad (\geq 0 \text{ by Harris}),$$

$$\Theta_i = b_i \Big[\sum_{j \in N_i} \sum_{k \in R_i \cap \Gamma(j)} D(B_j B_k) + \sum_{j < j' \in N_i} D(B_j B_{j'}) \Big], \qquad \Theta'_i = \sum_{j \in N_i} \sum_{k \in R_i \cap \Gamma(j)} D(B_i B_j B_k) + \sum_{j < j' \in N_i} D(B_i B_j B_{j'}).$$

Theorem A.1.

$$\prod_i \Big(1 - \min\Big\{b_i,\ b_i - C_i + \frac{\Theta'_i}{1-\beta_i}\Big\}\Big) \ \leq\ p \ \leq\ \prod_i \Big(1 - \max\Big\{0,\ b_i - \frac{C_i + \Theta_i}{1-\beta_i}\Big\}\Big).$$

Consequently, with $M_2 = \prod_i (1 - b_i + C_i)$, $|\log p - \log M_2| \leq E_3 := \sum_i \frac{\Theta_i + \Theta'_i + \beta_i C_i}{(1-\beta_i)(1-b_i)}$.

Proof. Fix i, write $N = N_i$, $R = R_i$, $F_X = \bigwedge_{j \in X} \neg B_j$ and $\nu = D(\cdot \mid F_R)$. Then $p = \prod_i (1 - c_i)$ with $c_i = \Pr[B_i \mid F_N \wedge F_R]$. Since S_i is disjoint from the supports of the events in R, $\nu(B_i) = b_i$ and

$$c_i = b_i - \frac{b_i\,[\nu(F_N) - \nu(F_N \mid B_i)]}{\nu(F_N)}, \qquad \nu(F_N) - \nu(F_N \mid B_i) = \nu\big(\textstyle\bigcup_N B_j \mid B_i\big) - \nu\big(\textstyle\bigcup_N B_j\big).$$

We use six facts. Each follows from the Harris inequality (an increasing and a decreasing event are negatively correlated under a product measure) and from independence of events with disjoint supports. For $j \in N$ let $R' = R \cap \Gamma(j)$ and $R'' = R \setminus R'$; the events of R'' have supports disjoint from $S_i \cup S_j$, and $\Pr[F_R] \leq \Pr[F_{R''}]$.

(H1) $\nu(B_j \mid B_i) \le D(B_iB_j)/b_i$ (Harris for B_iB_j versus F_R).

(H2) $\nu(B_j) \le b_j$ (Harris).

(H3) $\nu(B_j) \ge b_j - \sum_{k\in R'} D(B_jB_k)$: $\Pr[B_jF_R] \ge \Pr[B_jF_{R''}] - \sum_{k\in R'} \Pr[B_jB_kF_{R''}] \ge (b_j - \sum_k D(B_jB_k))\Pr[F_{R''}]$ by independence and Harris; divide by $\Pr[F_R] \le \Pr[F_{R''}]$ (trivial if the bracket is negative).

(H4) $\nu(B_jB_{j'}) \le D(B_jB_{j'})$ (Harris).

(H5) $\nu(B_j \mid B_i) \ge [D(B_iB_j) - \sum_{k\in R'} D(B_iB_jB_k)]/b_i$: as (H3), with B_iB_j in place of B_j.

(H6) $\nu(B_jB_{j'} \mid B_i) \le D(B_iB_jB_{j'})/b_i$ (Harris).

Upper bound on p. By Bonferroni, $\nu(\bigcup B_j \mid B_i) \le \sum_j \nu(B_j \mid B_i)$ and $\nu(\bigcup B_j) \ge \sum_j \nu(B_j) - \sum_{j<j'} \nu(B_jB_{j'})$. With (H1), (H3), (H4), $b_i[\nu(F_N) - \nu(F_N \mid B_i)] \le C_i + \Theta_i$; with (H2), $\nu(F_N) \ge 1 - \beta_i$. So $c_i \ge b_i - (C_i + \Theta_i)/(1 - \beta_i)$, and $c_i \ge 0$.

Lower bound on p. By Bonferroni the other way with (H5), (H6), (H2), $b_i[\nu(F_N) - \nu(F_N \mid B_i)] \ge C_i - \Theta'_i$. If $C_i \ge \Theta'_i$ use $\nu(F_N) \le 1$; otherwise use $\nu(F_N) \ge 1 - \beta_i$. In both cases (using $C_i \ge 0$) $c_i \le b_i - C_i + \Theta'_i/(1 - \beta_i)$; and $c_i \le b_i$ by Harris.

The E_3 bound. Put $x_i = 1 - b_i + C_i \ge 1 - b_i$. On the upper side, the factor is at most $x_i + (\Theta_i + \beta_iC_i)/(1-\beta_i)$ (or 1, in which case $1 - x_i < (\Theta_i + \beta_iC_i)/(1-\beta_i)$), and $\log(x+y) - \log x \le y/x$. On the lower side, the factor is at least $\max\{1 - b_i,\ x_i - \Theta'_i/(1 - \beta_i)\}$; if the second term is the larger one, $\log(x - y) - \log x \ge -y/(x - y) \ge -y/(1 - b_i)$, and otherwise $C_i < \Theta'_i/(1 - \beta_i)$ and $\log(1 - b_i) - \log x_i \ge -C_i/(1 - b_i)$, which is no smaller. Summing over i gives the claim. □

We checked the theorem against exact enumeration on 382 random instances (3–8 random monotone events, each a monotone DNF on 2–4 of 8–14 variables, random product distributions and orders; `experiments/check_second_order.py`): no violations, median $E_3 = 0.064$. The min in the lower bound matters: without it a factor can become negative, and a product of two negative factors can exceed p.

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